
Hidden Measurement Error in LLM Pipelines Distorts Annotation, Evaluation, and Benchmarking

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Abstract

LLM evaluations drive which models get deployed, what safety standards get adopted, which research conclusions get published, and how projections of AI’s labor-market impact get made. Yet standard confidence intervals ignore variability from judge model choice, model temperature, and prompt phrasing, producing under-coverage that worsens with more data. The omitted variance can shift results enough to reverse conclusions [Baumann et al., 2025, Huang et al., 2026]; pipelines that fail to average over it leave the surface that “benchmark hacking” exploits [Singh et al., 2025]. This paper decomposes LLM pipeline uncertainty into its sources, distinguishes variance that shrinks with more data from sensitivity to researcher design choices, and uses design-study projections to reduce total evaluation error (TEE). Across the demonstrations, naive standard errors are 40 - 60% smaller than the TEE-corrected SE. Using Chatbot Arena data, we show naive 95% CI coverage drops as n grows while TEE-corrected coverage holds at 95%, and TEE-guided pipelines restrict the benchmark gaming surface from 56 to 32 Elo ($K = 27$), below the human-leaderboard baseline. We show further that a small pilot recovers honest CIs and projects which design changes most improve precision. Acting on those projections roughly halves MMLU estimation error against the answer key at equivalent cost, and raises per-match agreement with human votes by 7.9 percentage points on Chatbot Arena.

1 Introduction

LLM measurement pipelines introduce variance at multiple stages that traditional reporting norms ignore. Researchers who use LLMs for annotation, evaluation, or content moderation routinely do so with no assessment of prompt sensitivity, temperature dependence, or item-level heterogeneity [Pecher et al., 2026, Barrie et al., 2025]. Benchmark scores likewise depend on prompt wording and scoring model choice, and this dependence is seldom measured (SI Appendix, Table 1).² In all cases, unmeasured variation can create analogues to p -hacking [Simmons et al., 2011, Gelman and Loken, 2013] and overfitting [Recht et al., 2019] such that the results reflect measurement noise, not general phenomena.

At best, current practice reports confidence intervals (CIs) reflecting variation from repeated sampling at identical settings, ignoring prompt sensitivity, item heterogeneity, and their interactions. Figure 1 presents simulation results showing that we should expect under-coverage to *worsen* with more items, because omitted variance from prompt and judge choices does not shrink with N .³

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²AlpacaEval and Chatbot Arena report item-sampling CIs; AILuminare uses a multi-judge evaluator ensemble; no benchmark decomposes prompt, temperature, or interaction variance. Full 13-benchmark survey in SI Appendix, Table 1.

³Variance parameters estimated via REML under the framework introduced below; 1,000 simulation replicates per data point (SI Appendix, Section E.4).

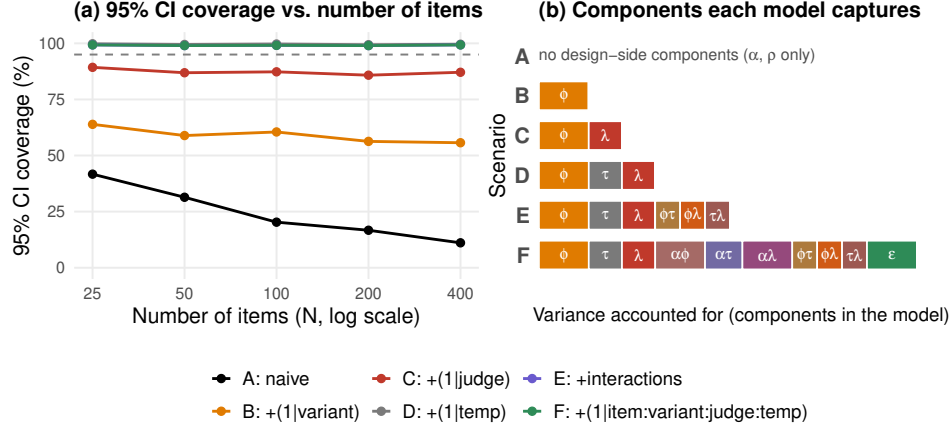


Figure 1: Naive confidence intervals fail because pipeline variance from prompt, judge, and temperature choice does not shrink with items (N). (a) 95% CI coverage versus N in six scenarios (A - F). (b) Variance components each scenario accounts for. Scenario A is the naive estimate $\Phi^{-1}(0.975) \times s/\sqrt{N}$; B - F add variance parameters from the framework below, F including the item interactions and σ_ϵ^2 . D - F overlap near 99% (E lies under F).

The problem is especially urgent in benchmarking, where unstable scores make rank-order differences between closely spaced models (or systems under test, SUT) uninterpretable for evaluators and give model developers exploitable degrees of freedom. On stable measurement, Romanou et al. [2026] show that semantics-preserving input perturbations explain near half of performance variance, minor reorderings of answer choices shift MMLU leaderboard rankings by up to 8 positions [Alzahrani et al., 2024], and on Chatbot Arena dropping just 0.003% of human preferences is enough to flip the top-ranked model [Huang et al., 2026]. With respect to exploitation, Singh et al. [2025] document best-of- K leaderboard gaming in production, the mechanism Section 4.4 quantifies.

Ignoring this variability can harm the scientific literature as well. One re-evaluation study found 31% of hypotheses tested with LLM annotations yielded incorrect conclusions under LLM configuration variation (model, prompt, temperature, output mapping), and found malicious actors could reverse the sign of 68% of statistically significant effects through adversarial configuration selection [Baumann et al., 2025]. Other work has found that formatting choices alone produced a 76-point accuracy spread [Sclar et al., 2024], while infrastructure nondeterminism shifted results by up to nine percentage points even at temperature zero [Yuan et al., 2025].⁴

2 Related Work

Several concurrent works address pieces of this problem, including variance decomposition [Haase et al., 2026, Wang, 2025], uncertainty quantification [Longjohn et al., 2025], sensitivity taxonomies [Romanou et al., 2026], meta-evaluation of benchmark reliability [McGregor et al., 2025], and tests of whether LLMs can replace human annotators [Calderon et al., 2025]. Camuffo et al. [2026] build a five-part variance taxonomy for LLM annotation and tie annotation variance to downstream bias. NIST AI 800-3 [Keller et al., 2026] fits a GLMM to benchmark data and separates benchmark accuracy from generalized accuracy as estimands. Song et al. [2025] apply G-theory to writing assessment. Dobriban [2025] note that a unified statistical methodology for evaluation “remains to be developed” (SI Appendix, Section C.9).

3 Framework

We call the framework *Total Evaluation Error* (TEE, Figure 2). TEE decomposes pipeline variance into components that include the first-order interactions, then projects from those components the standard error of any design a researcher might run. The projections are what let us calibrate CIs,

⁴ $T = 0$ does not remove model uncertainty. Greedy decoding returns one deterministic path, but the token distribution keeps its entropy, and for chain-of-thought models early token choices compound along that path [Renze and Guven, 2024].

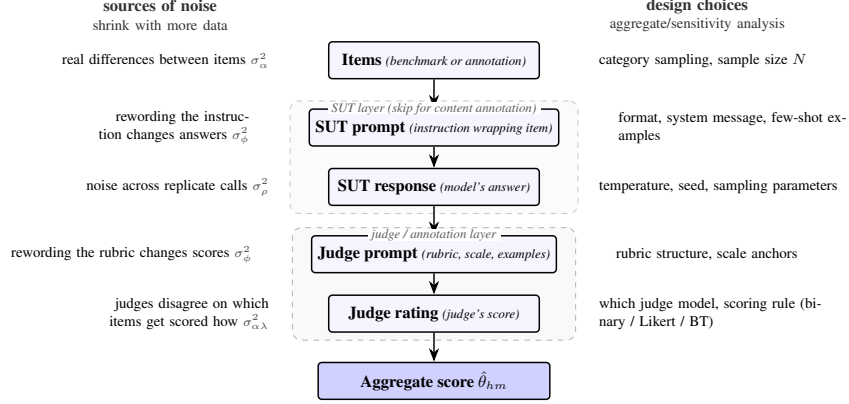


Figure 2: TEE separates aggregation-reducible noise from design-choice sensitivity at every pipeline stage. For content-annotation tasks (e.g., labeling social media posts) the SUT layer is omitted.

allocate budget, and shrink the surface that benchmark gaming exploits. The name follows total survey error [Sen et al., 2021, Groves et al., 2010], which treats every stage of a measurement pipeline as a source of error, and the framework draws on the text-as-data literature [Grimmer and Stewart, 2013].

3.1 Relation to Generalizability Theory

TEE operationalizes generalizability theory (G-theory) for LLM pipelines [Cronbach et al., 1972, Brennan, 2001, Shavelson and Webb, 1991, Bayerl and Paul, 2007] and uses its standard estimators. Our contribution is the LLM-specific measurement design that makes prompts, temperatures, models, judges, scoring methods, and interactions such as item $\times$ judge and prompt $\times$ judge explicit sources of pipeline error, where today they are hidden implementation choices. Four links to G-theory carry the rest of the paper.

(1) The universe of admissible observations here spans items, prompt phrasings, judge models, SUT models, decoding temperatures, replications, and scoring protocols. The object of measurement is the *item*, the unit whose score the evaluation generalizes over. On MMLU an item is a benchmark question. On safety and Arena a judge scores model output, and an item is the fixed evaluation case, a user prompt plus the SUT response(s) being judged. The prompt variants that instruct the judge index a facet of their own. (2) We apply the random versus fixed distinction to LLM design choices. Facets we sample over (items, prompt variants, replications) contribute uncertainty that averaging shrinks. A model, temperature, or scoring protocol we hold fixed is a design feature, and its contribution belongs in a sensitivity analysis (Section 3.2). (3) A G-study diagnoses which facets and interactions dominate under the design actually run (Section 3.3). A D-study projects which alternative design would reduce the target SE (Section 3.4). We map those projections onto budget choices and test how far the rankings survive pilot-size limits and misspecification. (4) The scoring protocol produces the observed score, so it belongs to the measurement design. Estimated components change under binary or Likert judging versus Bradley-Terry pairwise scoring, and on the probability versus the logit scale (SI Appendix, Sections H and G).

Classical G-theory grew up where raters are scarce and expensive while items and occasions are cheap, so its D-studies mostly ask how many raters one can afford [Brennan, 2001]. LLM pipelines price every facet in the same unit. An extra judge, prompt variant, item, or replication adds API calls in proportion to the rest of the design, and none of them needs recruiting or training, so every facet competes for one budget and the D-study has to rank them. TEE measures reliability. Judge correctness and construct coverage are validity questions that need external data, and Sections 4.1 and 6 say what to pair the decomposition with.

Figure 3(a) illustrates the TEE decomposition. Three LLM judges [Zheng et al., 2023] classify 141 safety items from the AILuminate benchmark. Judge design sensitivity contributes 43.8% of the variance with item $\times$ judge adding another 16.7%. On the other hand, prompt wording contributes less than 0.5%. In this example, investing in additional judges has higher returns to variance reduction than better prompts or additional benchmark items.

3.2 Data-Generating Process

An LLM measurement pipeline produces a scored output $Y_{ivhm}^{(r)}$ depending on five factors: item i ($i = 1, \dots, N$), prompt variant v ($v = 1, \dots, V$), temperature h ($h = 1, \dots, H$), model m (a judge at the judge layer, a SUT at the SUT layer), and replication r ($r = 1, \dots, R$). A replication is a repeated API call with identical inputs. The framework specifies the linear mixed model:

$$Y_{ivhm}^{(r)} = \mu + \alpha_i + \phi_v + \tau_h + \lambda_m + (\alpha\phi)_{iv} + (\alpha\tau)_{ih} + (\phi\tau)_{vh} + (\alpha\lambda)_{im} + (\phi\lambda)_{vm} + \epsilon_{ivhm} + \rho_{ivhm}^{(r)} \quad (1)$$

where μ is the grand mean, α_i is the item random effect, ϕ_v the prompt variant random effect, τ_h the temperature fixed effect, λ_m the model fixed effect, the five parenthesized interaction terms are two-way interaction random effects, $\epsilon_{ivhm} \sim N(0, \sigma_\epsilon^2)$ captures cell-level error from unmodeled higher-order interactions, and $\rho_{ivhm}^{(r)} \sim N(0, \sigma_\rho^2)$ captures within-cell replicate noise.

We treat items and prompt variants as *random* because we wish to get an estimand over all items and prompts that might have been drawn, so we average over them. On the other hand, we treat temperature and model as *fixed* because we purposefully assembled the judge panel and do not wish to generalize to some model population, and similarly we consider temperatures as analogous to a “dose” rather than something to be sampled. Temperature also has nonlinear, non-additive effects on output variance [Renze and Guven, 2024, Li et al., 2025a] (SI Appendix, Section O).

3.3 G-Study: Estimating the Variance Components

The G-study is the factorial we actually run, and its output is one variance component per facet and interaction under that observed design. Three assumptions hold the decomposition together. The first is exchangeability. Any repeated call at the same settings could stand in for any other, and no prompt variant is systematically better than the rest. That is what makes prompts and replications draws from a population, so their contributions shrink as a design adds variants or repeated calls. A prompt that reliably beats its alternatives is a design choice, and adding variants would only dilute it. The second is additivity up to two-way interactions. Each variance component then belongs to one or two facets, and averaging over them divides its contribution by their counts. Higher-order terms land in the residual, which does no harm so long as they stay small. The third is that random effects are independent and normally distributed. Independence lets the total variance equal the sum of its parts. If the difficult items were also the prompt-sensitive ones, a cross term would appear that no facet owns. Normality does less work. It supplies the likelihood and the 1.96 in the confidence interval, and the variance estimates stay consistent without it [Jiang, 1996]. We state each assumption formally in SI Appendix, Section C, and test how D-study projections hold up when each is violated in Section E.

Which variants are admissible is therefore a construct question, and we settle it with a screen that never sees scores [Cronbach et al., 1972, Groves et al., 2010, Mizrahi et al., 2024, Sclar et al., 2024, Barrie et al., 2024]. We anchor on the original prompt and require each candidate to preserve the task or rubric, the inserted content, the answer format, and the reasoning and scoring mode. A deterministic check rejects candidates that drop an input placeholder or the answer-format instruction. An independent screener model, never used as a judge, then rates construct equivalence from 1 to 5, and we reject anything below 4. On safety and MMLU the decoys score 1.0 and 2.5, and every retained variant scores 4 or 5. Applied retroactively, the deterministic check flags one MMLU variant that the screener had passed, and we drop it (Section 4.2). We then pick diverse admissible variants and report how the estimates move with breadth. Widening from strict paraphrase to the framing variants raises the MMLU prompt-attributable share of per-observation G-study variance from 2.1% to 7.3%, and a broader hand-authored set reaches 20.6%, so we do not claim coverage saturation there. The safety share holds at 1.3% across narrow, broad-wording, and broad-framing sets (SI Appendix, Section K). These per-observation shares are a different quantity from the $\text{Var}(\hat{\theta})$ shares reported in Section 4.

At fixed temperature h and model m , the observation-level variance decomposes as:

$$\begin{aligned} \text{Var}(Y \mid h, m) = & \sigma_\alpha^2 + \sigma_\phi^2 + \sigma_{\alpha\phi}^2 + \sigma_{\alpha\tau}^2 \\ & + \sigma_{\phi\tau}^2 + \sigma_{\alpha\lambda}^2 + \sigma_{\phi\lambda}^2 + \sigma_\epsilon^2 + \sigma_\rho^2 \end{aligned} \quad (2)$$

While the decomposition conditions on the temperature and model main effects (τ_h and λ_m), the interaction terms remain because they are variance components of random effects. When averaging over M models, a design-sensitivity term $\sigma_\lambda^2 = \frac{1}{M} \sum_m (\lambda_m - \bar{\lambda})^2$ is added. The same is true for averaging over multiple temperatures. Equation 1 omits the temperature $\times$ model interaction. A design that averages over both factors also adds a term for their interaction, $\sigma_{\tau\lambda}^2$ (SI Appendix, Section C).

In practice the G-study is a single lmer call [Bates et al., 2015]. Fixed factors enter as covariates, temperature + model, and every random facet or interaction enters as its own term, for example (1|item) + (1|prompt_variant) + (1|item:model). REML [Jiang, 1996] returns one variance estimate per term, and those estimates are the σ^2 components in Equation 2. The D-study below divides each component by the number of levels of its facets that a proposed design averages over. The full formula, pilot-design thresholds, and bootstrap-CI guidance for small facet counts are in SI Appendix, Sections C.6 and M.

3.4 D-Study: Projecting Precision Under Alternative Designs

The G-study reports which facets dominate under the design we ran. A D-study takes the estimated variance components and projects the variance of the grand mean $\hat{\theta}$ under any design a researcher might run. Following G-theory, primes mark the proposed design (N' items, V' prompt variants, R' replications) and unprimed letters the estimation design (the G-study). Each component enters the projection divided by the number of levels the proposed design averages it over. Item heterogeneity is divided by N' , prompt sensitivity by V' , and judge sensitivity by the panel size M . Each interaction is divided by the product of the counts for the facets it spans, cell-level error by $N'V'M$, and replicate noise by $N'V'MR'$. The formula treats items as draws from a larger population. When a benchmark's fixed item set is itself the target, the σ_α^2/N' term drops out.

The item $\times$ judge and prompt $\times$ judge terms, $\sigma_{\alpha\lambda}^2$ and $\sigma_{\phi\lambda}^2$, are identifiable only when the estimation design crosses M models; with a single model they fold into σ_α^2 and σ_ϕ^2 (SI Appendix, Section C).⁵ The cell-level and replicate-noise variances are estimated per model and averaged over the panel, $\bar{\sigma}_\epsilon^2 = \frac{1}{M} \sum_m \sigma_{\epsilon,m}^2$ and $\bar{\sigma}_\rho^2 = \frac{1}{M} \sum_m \sigma_{\rho,m}^2$. At fixed temperature, the projected variance is

$$\text{Var}(\hat{\theta}_h) = \frac{\sigma_\alpha^2}{N'} + \frac{\sigma_\phi^2}{V'} + \frac{\sigma_\lambda^2}{M} + \frac{\sigma_{\alpha\phi}^2}{N'V'} + \frac{\sigma_{\alpha\tau}^2}{N'} + \frac{\sigma_{\phi\tau}^2}{V'} + \frac{\sigma_{\alpha\lambda}^2}{N'M} + \frac{\sigma_{\phi\lambda}^2}{V'M} + \frac{\bar{\sigma}_\epsilon^2}{N'V'M} + \frac{\bar{\sigma}_\rho^2}{N'V'MR'}$$

The D-study variance gives the standard error and 95% CI: $\hat{\theta} \pm 1.96 \cdot \sqrt{\text{Var}(\hat{\theta})}$, accounting for all identified pipeline variance. When facet counts are small (e.g., $V = 3$), the result is moderate undercoverage (80 - 90% instead of 95%). Parametric bootstrap estimates restore nominal coverage at minimal additional cost (SI Appendix, Section E.4).

In Monte Carlo simulations (1,000 replicates), the D-study picks the correct top intervention in 98% of runs under the default DGP. Under four assumption violations (correlated random effects, non-exchangeable prompts, non-normal scores, heterogeneous category interactions), projected variance stays within 8% of the truth. A latent-ambiguity sweep that adds sparse three-way interactions keeps component-level bias below 9% (SI Appendix, Sections E.2 and E.5).

Choosing facets and judges. Do not cross everything. Pilot over the factors that actually vary in deployment, fit the decomposition, and spend where the D-study ranking shows non-trivial variance. In our demonstrations one or two factors dominated, usually judge choice, and temperature and extra replications rarely mattered. Sampling parameters such as top- p or top- k enter exactly as temperature does. Direct versus chain-of-thought prompting, output format, and reasoning mode can change the construct itself, so model them as separate factors unless they pass the construct-equivalence screen of Section 3.3. For judges, begin with a plausible and diverse panel and run a small crossed pilot. Judge marginals and leave-one-judge-out refits then sort calibration offsets from boundary-case disagreement (Section 4.1), and human labels settle the question when they

⁵When items belong to categories, as in the demonstrations here, $\sigma_\alpha^2 = \sigma_\kappa^2 + \sigma_\delta^2$ splits item heterogeneity into between-category and within-category parts. Sampled categories would add a σ_κ^2/C term. The categories here are fixed (specific hazard types, ideology dimensions, MMLU subjects), so σ_κ^2 stays inside σ_α^2/N' .

exist. Different prompt or judge selections can change the recommendation, so TEE is a diagnostic for the pipeline in hand.

4 Empirical Demonstrations

We apply the TEE decomposition across three empirical domains in the main text: binary safety classification (AILuminate), multiple-choice QA (MMLU), and pairwise model comparison validated against human votes (Chatbot Arena). On safety and Arena the model facet is the judge panel. On MMLU there is no judge, and the model facet is the SUT scored against the answer key (Section 3.1). Section 4.4 then quantifies the gaming surface single-configuration benchmarks expose. In the SI, we provide a political affinity annotation demonstration (Section I.1) with a 9-annotator human baseline, a Likert vs pairwise scoring comparison (Section H), and a pilot validation showing that a small sample recovers the qualitative D-study guidance of a $4.5\times$ larger full run (Section I).

4.1 Safety: Judge Disagreement Dominates

We first illustrate the variance decomposition using AILuminate v1.0 [Ghosh et al., 2025], a safety benchmark from MLCommons. The benchmark assesses whether LLM responses cross safety lines across 12 hazard categories (hate speech, self-harm, violent crime, etc.). Most items are unambiguously safe (94% safe rate) and the variance concentrates on the 6% of borderline cases where judges disagree about whether a refusal is adequate. We decompose the variance across judges, prompts, temperatures, items, and their interactions, based on 36 DEMO prompts (3 per category) crossed with responses from Claude Opus 4.5, GPT-5.1, DeepSeek Chat v3.1, and Grok 4.1 Fast (141 items after dropping 3 SUT refusals), scored under 5 prompt variants $\times$ 3 temperatures $\times$ 3 judges $\times$ 8 replications = 50,760 calls.

The G-study (Figure 3(a)) identifies judge design sensitivity as the dominant lever, contributing 43.8% of $\text{Var}(\hat{\theta})$ at the operational design ($N=141$, $V=5$, $M=3$, $H=3$, $R=8$). Item $\times$ judge interaction adds another 16.7%. Judge choice shifts the headline rate, and panel members disagree about which specific items are safe. Aggregating across the three judges in the panel cuts the safety SE by 38% because the model design-sensitivity σ_{λ}^2 and the item $\times$ judge interaction $\sigma_{\alpha\lambda}^2$ both shrink as $1/M$ when the projection averages over M judges (Eq. 3).

The three judge-related terms (judge design sensitivity, item $\times$ judge, prompt $\times$ judge) sum to 73% of $\text{Var}(\hat{\theta})$. To check that no single judge produces this, we refit the model dropping each judge in turn, and the judge-related share stays between 68% and 83% (SI Appendix, Section R). The decomposition cannot say which judge is right. It does say where to look. A judge whose mean sits away from the rest of the panel calls for recalibration or removal, and items with large item $\times$ judge disagreement are the boundary cases that need substantive review against human labels, known-answer items, or the hazard taxonomy.

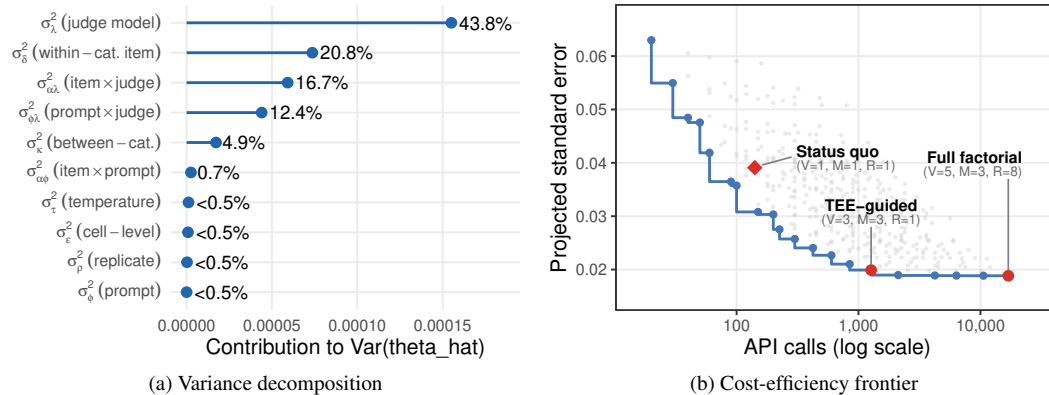


Figure 3: Safety decomposition (a) and cost-efficiency frontier (b) for AILuminate. Judge-model variance dominates the decomposition in (a). A TEE-guided design that aggregates over diverse judges captures most of the available SE reduction at modest cost (b).

Within-category item heterogeneity contributes 20.8% and prompt $\times$ judge 12.4%, with each remaining component below 5%. Replicate noise σ_{ρ}^2 and cell-level error σ_{ϵ}^2 are large components at the

item level, but enter $\text{Var}(\hat{\theta})$ divided by $N'V'H'M'R'$ and $N'V'H'M'$ respectively, so each contributes less than 0.5% to the variance of the mean. Temperature and prompt main effects contribute negligibly.⁶

The D-study then projects precision under any proposed design (any new combination of $N'V'H'M'R'$, etc), enabling a cost-efficiency analysis. Figure 3(b) plots projected standard error against API cost for 360 candidate designs of the safety benchmark. The status quo (141 items, one prompt, one judge, no replication) sits well above the efficient frontier; at the same cost, frontier designs that trade items for judge diversity achieve 9% lower SE. A TEE-guided estimation strategy (141 items, 3 prompts, 3 judges, no replication) cuts SE by 49% at $9\times$ the cost, while the full factorial (5 prompts, 3 judges, 8 replications) provides only an additional 5% reduction at $120\times$ the cost. The frontier flattens beyond approximately 1,000 calls, indicating steep diminishing returns to additional investment.

4.2 MMLU: Item and Prompt Reallocation Roughly Halves Error

Where LLM-as-judge is at the center of the safety demonstration, our MMLU benchmark analysis removes the judge layer entirely. MMLU (Massive Multitask Language Understanding) [Hendrycks et al., 2021] provides ground-truth labels for multiple-choice questions, letting us audit both pipeline accuracy and confidence-interval coverage against those answers. Three SUTs answered a 200-item pool under 4 prompt variants, 3 temperatures, and 8 replications (SI Appendix, Section J).

The G-study (Figure 4(a)) attributes most of $\text{Var}(\hat{\theta})$ to which questions are sampled. Within-category item heterogeneity contributes 72.2% and between-category heterogeneity 7.7%, followed by item $\times$ SUT (10.5%) and SUT-model design sensitivity (6.7%). Prompt wording barely matters on MMLU. The three prompt-related terms together hold 2.7% of $\text{Var}(\hat{\theta})$, and accuracy moves only from 83.9% to 85.3% across the four prompt variants.⁷ The logit-link refit agrees, with item heterogeneity leading at 71.4% (SI Appendix, Section G).

The D-study then frames the evaluation as a budget-allocation problem, asking how to split a fixed total call budget across items (N), prompt variants (V), and replications (R). A 30-item pilot supplies the variance components, and the remaining 170 items form the pool. We compare three allocations, with the *naive* allocation ($V = 1$, $R = 3$) replicating each call three times, the *standard* allocation ($V = 1$, $R = 1$) skipping replication to fund more items, and the *TEE-guided* allocation letting the D-study decide between adding items, prompts, or replications at each budget level. Because the two item terms hold 80% of $\text{Var}(\hat{\theta})$, the D-study buys items first, so TEE and standard coincide until the pool runs out. Naive spends two-thirds of its calls on replications, which barely reduce error, and its RMSE is 1.6 to 2.4 times TEE’s at every budget (Figure 4(b)). Once all 170 items are in use, standard stops improving, while TEE averages over prompt variants and roughly halves standard’s RMSE at the largest budgets. TEE’s 95% CIs cover at 97 - 100% throughout, and naive CIs under-cover only at the smallest budgets (91% and 93%).

4.3 Chatbot Arena: Coverage, Accuracy, D-Study Validation

Chatbot Arena [Chiang et al., 2024] extends our empirical tests to data comprising pairwise human votes that serve as ground-truth labels. LLM judges can stand in for human voters only if their scores hold up to the same standards as a human-rater study, with accuracy that does not depend on idiosyncratic prompt or judge choices. We test a TEE-guided pipeline against a naive single-judge baseline on 4,676 Arena matches drawn from the public release `lmarena-ai/arena-human-preference-100k`, stratified by (model $\times$ category) across Creative Writing, Coding, Factual QA, and Persuasion. Each match was scored under five chain-of-thought prompt variants by three judges, in both Likert (1 - 5 per response) and pairwise (A / B / tie, matching Chatbot Arena’s voting UI) modes; full design details in SI Appendix, Section T.

⁶Three-way interaction terms (item $\times$ prompt $\times$ temperature, item $\times$ temperature $\times$ judge) are statistically significant but substantively negligible, supporting the two-way interaction specification.

⁷A fifth prompt variant gave the model only the question, the four options, and “Answer:”, with no instruction to reply with a letter. With responses capped at 8 tokens, models often began explaining their reasoning and were cut off before choosing an option. Counting those cut-off responses as wrong would give the prompt-related terms 70% of $\text{Var}(\hat{\theta})$. That 70% measures a missing answer-format instruction. The admissibility check of Section 3.3 exists to catch exactly this, so we exclude that variant (SI Appendix, Section J).

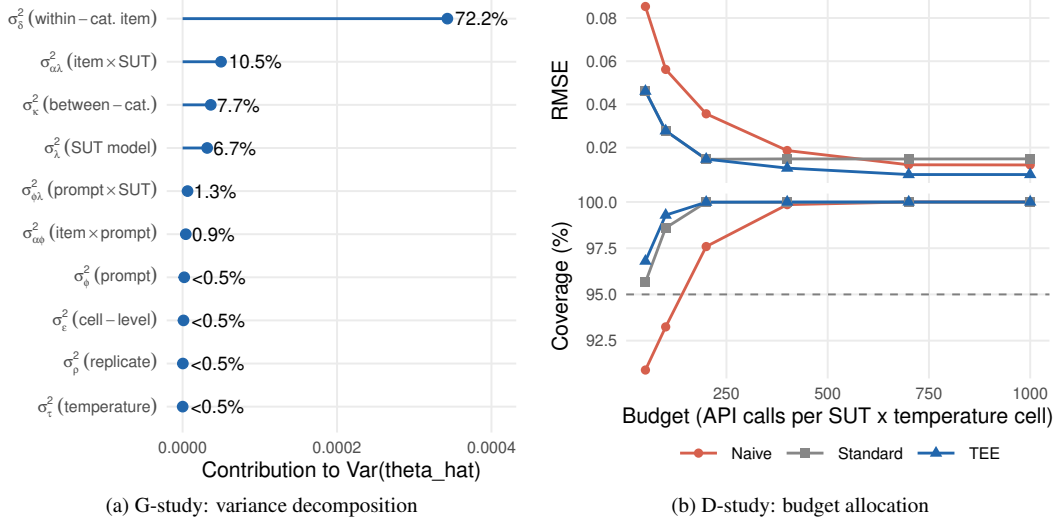


Figure 4: MMLU G-study (a) and D-study (b). (a) Contribution of each component to $\text{Var}(\hat{\theta})$ at the operational design ($N=200$, $V=4$, $M=3$ SUTs, $H=3$, $R=8$); item heterogeneity dominates, and the prompt terms together hold 2.7%. (b) RMSE against ground-truth accuracy (top) and 95% CI coverage (bottom) by budget. TEE-guided allocation matches standard until the 170-item pool is used up, then keeps improving by averaging over prompt variants; naive, which replicates each call three times, has 1.6 to 2.4 times TEE’s RMSE. 1,000 Monte Carlo replicates in each of the 9 SUT $\times$ temperature cells at every budget level.

A Monte Carlo experiment that treats the full 4,676-match Likert factorial as a population ($p^* = 0.389$), shows that TEE averaging gives well-calibrated uncertainty. Naive 95% Wald CIs computed from a single (judge, variant) cell hold nominal coverage at $n_m = 100$ matches but drop to 78% coverage at $n_m = 2,000$ (Figure 5). The TEE crossed random-effects Wald CI sits at or above 95% throughout. The naive SE shrinks at $1/\sqrt{n_m}$ while between-cell variance does not, so at large n_m the naive interval becomes a tight band around a single cell’s mean. As in Figure 1, adding cases widens the coverage gap.

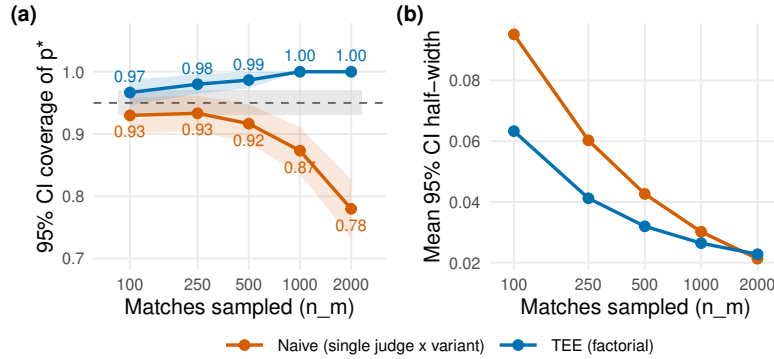


Figure 5: Naive coverage on Chatbot Arena drops as the sample grows, while TEE maintains nominal coverage. (a) Empirical 95% CI coverage versus match sample size n_m for the naive single-judge, single-variant design and the TEE factorial design; ribbons show ± 1.96 Monte Carlo SE over 300 replicates. (b) Mean 95% CI half-width versus n_m . The naive half-width shrinks as $1/\sqrt{n_m}$ while TEE bottoms out at the floor $\sigma_V^2/V + \sigma_J^2/J$ set by variant and judge variance. The target $p^* = 0.389$ is the mean over the 15 observed judge $\times$ variant cells (3 judges, 5 variants), a finite-panel quantity. The TEE interval targets the superpopulation of prompts and judges from which those cells were drawn, so it carries between-cell variance the finite-panel target does not, and it over-covers that target at large n_m . An interval for the finite-panel target would hold the 5 variants and 3 judges fixed and bootstrap over matches only.

TEE averaging also improves per-match agreement with the individual human vote across all four categories (SI Appendix, Figure 20). Pooled across 4,676 matches, TEE Likert raises AB-only agreement over single-judge Likert by +7.9 pp (74.9% vs. 67.0%) and TEE pairwise raises it over single-judge pairwise by +3.6 pp (71.9% vs. 68.3%, see SI Appendix Table 22).

A D-study fit on 80% of matches projects per-category SE reductions of 0.53 - 0.59 from single-judge to TEE. Across the four held-out categories, projected SE reductions track observed Likert agreement lifts at Spearman $\rho = 0.80$ ($n = 4$; SI Appendix Figure 21). Both gains matter when matches are scored one at a time, as in training on Arena labels [Egami et al., 2023].

4.4 Single-Configuration Benchmarks Vulnerable to Benchmark Hacking

Singh et al. [2025] document Meta running 27 private Llama-4 variants on Chatbot Arena before launch and publishing only the top scorer (*benchmark hacking*), which constitutes best-of- K exploitation of pipeline noise. This can play out in a few ways: on hosted leaderboards, developers can submit many model variants under different identifiers and report only the best-scoring submission. On open-source benchmarks, they can rerun the evaluation internally with different prompt formats or scoring configurations until the score is favorable. The expected inflation is $E[\max(Z_1, \dots, Z_K)] \cdot \sigma_{\text{pipeline}}$, where σ_{pipeline} is the standard deviation of the pipeline noise decomposed above. Prompt formatting alone can produce up to 76 accuracy points of spread on identical models [Sclar et al., 2024], suggesting the gameable surface is broad in practice.

We quantify the gaming surface in the context of Chatbot Arena by bootstrapping 4,676 matches across pipeline configurations (Figure 6). The Arena human leaderboard already exposes a 45-Elo surface at $K=27$ from voter sampling alone, large enough to flip rankings between adjacent models. A single-configuration LLM-judge replacement ($V=1, M=1$) increases the surface to 56 Elo, since a developer can target both the LLM-judge configuration and the underlying voter sample. Averaging across either prompts or judges alone recovers the human baseline ($V=5, M=1$: 46 Elo; $V=1, M=3$: 46 Elo). TEE-LLM ($V=5, M=3$ averaging) cuts the surface to 32 Elo, about 30% below the human leaderboard. Judge main effects account for the largest share of gameable LLM variance (Table 3), the component a developer most directly controls when picking judge models. Best-of- K is one facet of the manipulation surface; Singh et al. [2025] also document score retraction, disproportionate sampling of proprietary models (61.4% of battles), and Arena-specific data access that lifted Arena performance by up to 112% without external gains.

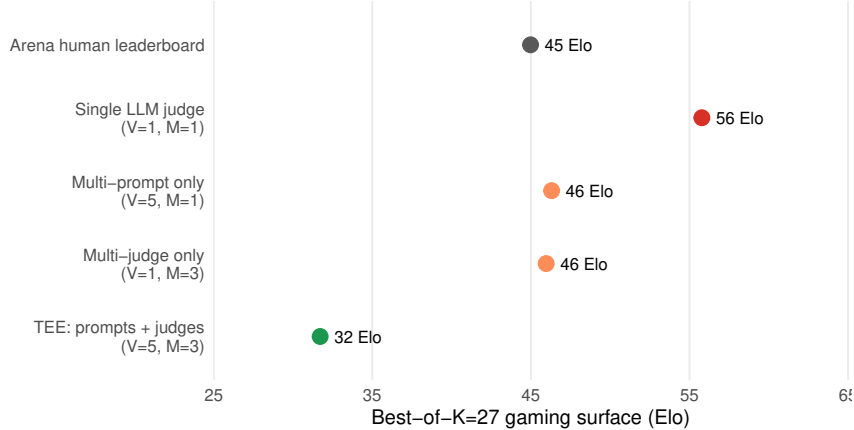


Figure 6: Best-of- $K=27$ gaming surface on Chatbot Arena across pipeline configurations. “Gaming surface” is the expected Elo boost a developer earns by submitting $K=27$ model variants and reporting only the highest-scoring one. Variance surface computed by bootstrapping 4,676 Arena matches.

5 Discussion

Across the demonstrations, naive error estimates understate measurement uncertainty because they omit judge disagreement, prompt sensitivity, replicate noise, and their interactions. Evaluators then

mistake pipeline noise for capability differences, and researchers mistake it for generalizable patterns in LLM-annotated data. The gap scales with the subjectivity and complexity of the task, and the Arena, MMLU, and propaganda results show that failing to average over these sources also costs accuracy.

Pipeline variance can also serve as a gaming surface in benchmarking and in research. We show that submitting best-of $K=27$ variants (as documented in Singh et al. [2025]) inflates Chatbot Arena Elo scores by 56 points, while averaging over multiple judges and prompts cuts that surface by 43% (§4.4). The same vulnerability extends to LLM-annotated research, where pipeline variance maps onto labeling error and predicts misclassification against the 9-rater human consensus in our propaganda demonstration ($r = -0.68$, SI Appendix, Section I.1), consistent with the configuration- and adversarial-search reversals Baumann et al. [2025] document across 37 LLM-annotation tasks. The implications extend beyond model capability overestimates, as benchmarks like SWE-Bench, LegalBench, and HealthBench inform estimates of labor substitution and productivity gains [Eloundou et al., 2023] associated with widespread LLM adoption.

What transfers across pipelines? The structural lessons do. Claims are almost always about an item population, so more items beat fewer, and no score should hinge on a single judge or a single phrasing. The empirical rankings travel as priors. Wherever a judge layer is present, averaging over judges yields the largest single reduction in our projections, with items next on AILuminate and prompts next on Arena. The budget-relevant magnitudes are local estimates, and a pilot in the target pipeline should replace them before anyone spends on their basis.

Pipeline calibration is upstream of all real world benchmark use. When that output drives safety certification, LLM-annotated research, and labor-market projections, underestimated uncertainty propagates into each downstream decision. Our TEE framework allows for calibrated CIs at a fraction of that downstream cost.

6 Limitations

TEE has a number of simplifying assumptions worth noting. When a model emits reasoning traces, the variance they generate is not attributed under the current framework and instead inflates σ_ϵ^2 . When the scoring function relies on an LLM-as-judge, its variance may compound with the SUT’s [Haldar and Hockenmaier, 2025, Chehbouni et al., 2025], and any SUT $\times$ judge interactions this introduces fall outside the within-layer decomposition (SI Appendix, Section Q). Variance shares are reported on the probability scale and therefore assume an LPM, which generally preserves qualitative rankings but not precise magnitudes compared with a logistic link function (SI Appendix, Section G).

Perhaps more importantly, TEE cannot say whether the items span the construct of interest, which of course limits generalizability. Total survey error has long called this coverage error [Sen et al., 2021], and a toxicity benchmark that omits coded hate speech will yield precise but domain-incomplete estimates.

Finally, all demonstrations are in English and use frontier models, leaving high-ambiguity tasks (creative writing, open-ended reasoning) untested. The full factorial design requires $\sim 54,000$ API calls, though a pilot subset can deploy much of the framework in practice.

7 Conclusion

Confidence intervals in LLM evaluation are systematically too narrow, and benchmarks reward optimization against measurement noise. Beyond diagnosis, TEE lets analysts run D-study projections from a small pilot to recover honest CIs and rank which design changes would most reduce noise (SI Appendix, Section I). As evaluations increasingly shape deployment decisions, safety certifications, regulatory standards, and projections of AI’s labor-market and economic impact, closing this gap is necessary before evaluation scores can support the high-stakes judgments they inform.

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A Quickstart Guide

This guide walks through the operational pipeline for diagnosing uncertainty in an LLM annotation workflow. The full framework and derivations appear in Section C.

1. **Run a small pilot.** A small pilot at a fixed temperature can estimate the dominant variance components when the judge layer is varied. The minimum design uses 30 or more items, 3 or more prompt variants, 2 or more judges (when a judge layer exists), and 3 or more replications, for roughly 540 API calls if evaluating a single scoring function. Variants share task instruction, output format, and reasoning mode but may differ in word choice, sentence structure, and framing of the construct (Section C, Assumption 1). This broad operationalization reflects the prompt-wording uncertainty a real evaluator faces. Both strict paraphrases (e.g., prompting Claude Haiku at temperature 0.9 to rewrite a seed instruction V ways) and small sets of templates that vary the framing of the construct are valid. The latter produces a broader $\hat{\sigma}_\phi^2$. Include the original benchmark prompt as one of the V variants where one exists. Items should be drawn from the target deployment domain. Drop the judge facet for benchmarks without one (e.g., MMLU).
2. **Fit the variance decomposition.** The `totalevalerror` R package (<https://github.com/SolomonMg/totalevalerror>) wraps the variance decomposition, D-study, and budget-allocation routines from this paper:

```
library(totalevalerror)
des <- tee_design(pilot_df,
  outcome = score, item = item_id,
  prompt = variant_id, judge = judge_model,
  temperature = temp, replicate = rep,
  scoring = "binary")
fit <- tee_decompose(des)
```

The fit reports each variance source as a share of $\text{Var}(\hat{\theta})$, and the largest share identifies the bottleneck. If judge variance (σ_λ^2) or item $\times$ judge ($\sigma_{\alpha\lambda}^2$) dominates, as in the safety demonstration, adding judges helps most; if within-cell replicate noise (σ_ρ^2) leads, more replications buy precision; if prompt sensitivity (σ_ϕ^2) dominates, more prompt variants help; if item heterogeneity (σ_α^2) leads, more items help. The underlying fixed-temperature lmer formula is $\text{score} \sim \text{judge} + (1|\text{item}) + (1|\text{prompt_variant}) + (1|\text{item}:\text{prompt_variant}) + (1|\text{item}:\text{judge}) + (1|\text{prompt_variant}:\text{judge}) + (1|\text{item}:\text{prompt_variant}:\text{judge})$, with the judge terms dropped for benchmarks without a judge layer. The final random effect captures cell-level idiosyncrasy; the lmer residual captures within-cell replicate noise. Users who prefer to fit it directly can do so.

3. **Optimize with the D-study.** `tee_dstudy(fit)` returns projected SEs across alternative production designs (varying replications, prompt variants, judges, scoring methods). `tee_budget_allocate(fit, budget = X)` returns the cost-minimizing allocation under a fixed API budget. The projection is qualitatively reliable at pilot scale (Section I). The rank ordering of interventions transfers to the full design even when component magnitudes shift.
4. **Deploy and estimate CIs.** Apply the chosen production design to the full corpus, then call `tee_se(fit)` for the corrected SE and `tee_theta_ci(fit, level = 0.95)` for a 95% CI on $\hat{\theta}$. Because the formula incorporates the components naive CIs leave out (prompt and judge variance, plus their interactions), coverage holds near nominal even as sample size grows.
5. **Report.** `tee_writeup(fit)` produces a paragraph summarizing the decomposition, the dominant variance components, the corrected CI, and the D-study-recommended design. For each scored measurement, also report the exact prompt text, model version, temperature, and number of replications. A tiered reporting checklist appears in Section M, Table 19.

B Uncertainty Reporting in Current LLM Evaluation Practice

Table 1 surveys uncertainty reporting across thirteen widely used LLM benchmarks and annotation studies. Two patterns stand out. In our 13-benchmark survey, most benchmarks report no measurement uncertainty at all. MMLU, the Open LLM Leaderboard, MT-Bench, SWE-bench, and ARC-AGI all provide only a point estimate. The four that do report CIs (AlpacaEval, HELM, Chatbot Arena, Arena-Hard) capture only item-sampling variance, leaving prompt sensitivity, judge variance, temperature, and interactions unmeasured. Contamination is also widespread. All six benchmarks with public test sets (MMLU, AlpacaEval, HELM, MT-Bench, Arena-Hard, SWE-bench) have documented training-data leakage, and even Chatbot Arena leaks historical votes back into training data despite its live-query design. AILuminator’s private Official set and ARC-AGI’s held-out tasks are the only contamination-resistant designs in the survey.

The bottom row shows TEE addressing every TEE-decomposition column simultaneously: prompt sensitivity, judge model and temperature, replications, interactions, item sampling, and the basic CI/SE indicator. The remainder of this section walks through each benchmark in detail.

Table 1: Uncertainty reporting in prominent LLM evaluations. SUT-layer columns capture variance in the system under test. Judge-layer columns capture variance in the scoring pipeline.

	SUT layer					Judge layer				
	Any CI/SE	Item sampling	SUT prompt	SUT temperature	SUT replications	Judge prompt	Judge model	Judge temperature	Interactions	Contamination
MMLU [Hendrycks et al., 2021]	-	-	-	-	-	n/a	n/a	n/a	-	High
Open LLM Leaderboard	-	-	-	-	-	n/a	n/a	n/a	-	Mixed
AlpacaEval [Dubois et al., 2024]	*	*	-	-	-	-	-	-	-	High
HELM [Liang et al., 2023]	*	*	-	-	-	n/a	n/a	n/a	-	High
Chatbot Arena [Chiang et al., 2024]	*	*	-	-	-	n/a	n/a	n/a	-	Medium
MT-Bench [Zheng et al., 2023]	-	-	-	-	-	-	-	-	-	High
Arena-Hard [Li et al., 2024]	*	*	-	-	-	-	-	-	-	High
SWE-bench [Jimenez et al., 2024]	-	-	-	-	-	n/a	n/a	n/a	-	High
ARC-AGI [Chollet, 2019]	-	-	-	-	-	n/a	n/a	n/a	-	Low
AILuminator [Ghosh et al., 2025]	(*)	-	-	-	-	-	(*)	-	-	Medium
Gilardi et al. [Gilardi et al., 2023]	-	-	n/a	n/a	n/a	-	-	-	-	n/a
Törnberg [Törnberg, 2025]	-	-	n/a	n/a	n/a	-	-	-	-	n/a
Ziems et al. [Ziems et al., 2024]	-	-	n/a	n/a	n/a	(*)	(*)	-	-	n/a
TEE	*	*	*	*	*	*	*	*	*	n/a

Deterministic scoring (exact match or test suite), no judge layer.

Public test set with documented training-data leakage [Xu et al., 2024].

Most subtests public. GPQA gated to reduce contamination.

Human voters, not an LLM judge layer.

Live queries, but historical votes published and prompts recur across months.

Bootstrap CI over items, captures item sampling only.

Public task set with documented solution leakage into training data.

Private held-out evaluation set with human-calibrated difficulty.

Paper specifies bounds from evaluator false-safe/false-unsafe rates. Ensemble of evaluators fixed. Public leaderboard displays grade only.

Public DEMO subset (1,200 prompts) and public Practice set (12,000), with private Official test set (12,000).

MMLU and the Open LLM Leaderboard. MMLU [Hendrycks et al., 2021] evaluates each model once on a fixed test set (14,079 items) with a single prompt format. For open-weight models with accessible logits, the standard implementation (EleutherAI lm-evaluation-harness) scores by comparing log-likelihoods of the full answer text for each of the four choices. For API-only models, the evaluation generates text and parses the answer letter, reintroducing temperature dependence and replicate noise. In both cases, no uncertainty is reported. The dominant missing uncertainty is item heterogeneity. On our 200-item factorial decomposition (SI Appendix, Section J), within-category item heterogeneity contributes 72.2% of $\text{Var}(\hat{\theta})$, item $\times$ SUT 10.5%, between-category heterogeneity 7.7%, and SUT-model design sensitivity 6.7% at the operational design. The three prompt terms together hold 2.7%. Sclar et al. [2024] document up to 76 accuracy points of spread from formatting choices alone on MMLU. Those formatting perturbations are distinct from the phrasing-and-framing variants used in this paper (Assumption 1). Different implementations of MMLU (original, HELM, HuggingFace) also use different prompt templates and produce different rankings for the same models [Alzahrani et al., 2024]. The Open LLM Leaderboard follows similar practice. Most of its subtests use public test sets, with GPQA gated to reduce contamination.

AlpacaEval. AlpacaEval 2.0 [Dubois et al., 2024] uses a single LLM judge (GPT-4 Turbo) with a single prompt to score model outputs against a reference set. It reports a standard error capturing item-sampling variance. Every other pipeline factor is held to a single value (one judge, one prompt, one temperature, no replications), with none decomposed. A different judge model would produce a different win rate, but judge variance (σ_λ^2) is not measured. The length-controlled variant adjusts for response length but not for pipeline configuration variance.

HELM. HELM [Liang et al., 2023] evaluates each model on each scenario with a fixed prompt format and runs 3 random seeds over different in-context example selections to increase stability. All prompts, predictions, and test items are publicly inspectable, which aids reproducibility but also makes HELM vulnerable to test-set contamination through training data leakage [Xu et al., 2024]. Scoring uses deterministic automated metrics (accuracy, BLEU, ROUGE, F1), not LLM judges, so the judge layer does not apply. SUT prompt sensitivity (one format per scenario) and temperature effects (fixed) both apply but are not measured.

Chatbot Arena. Chatbot Arena [Chiang et al., 2024] computes Bradley-Terry [Bradley and Terry, 1952] coefficients (labeled as Elo ratings) via maximum likelihood estimation with sandwich robust standard errors, collecting an average of roughly 8,000 votes per model. Prompt wording varies organically across users but is not controlled or decomposed. Voter heterogeneity (different humans rank differently) and SUT temperature (fixed per model by the provider) both apply but are not separated from sampling noise. The bootstrap CIs capture vote-sampling uncertainty but not the model-variant selection effects documented by Singh et al. [2025], where providers tested dozens of private variants and published only the highest-scoring one.

MT-Bench. MT-Bench [Zheng et al., 2023] evaluates SUTs on 80 multi-turn questions (2 turns each, 8 categories) scored 1 - 10 by GPT-4. The judge model version is not pinned in the codebase, creating a reproducibility concern as the alias resolves to different model snapshots over time. The evaluation runs one prompt template per item at judge temperature 0 with no replications and no CI. SUT temperature varies by category (0.0 - 0.7). The LMSYS team’s own later analysis found that only 22.6% of model pairs have non-overlapping CIs on MT-Bench, compared with 87.4% on Arena-Hard [Li et al., 2024], so MT-Bench scores cannot statistically distinguish most model pairs.

Arena-Hard. Arena-Hard-Auto [Li et al., 2024] evaluates SUTs on 500 challenging user queries drawn from Chatbot Arena, scored by GPT-4-Turbo in pairwise comparison against a baseline (GPT-4-0314). A two-game position-swap design mitigates position bias. Bootstrap CIs (100 resamples) capture item-sampling variance but not judge variance (single judge, single prompt template, single temperature = 0, no replications). Arena-Hard achieves 87.4% model-pair separability, the highest among automated benchmarks tested. The test set is public.

SWE-bench. The SWE-bench family [Jimenez et al., 2024] evaluates coding agents on real GitHub issues. SWE-bench Verified (500 human-validated Python tasks) was the standard until contamination was documented across all frontier models. SWE-bench Pro (1,865 tasks drawn from 41 repositories, requiring ≥ 10 lines and averaging 4.1 files per fix) was introduced as a harder, less contaminated alternative. Scoring is deterministic. The agent’s patch is applied and the repository test suite is run. Pass@1 (single attempt) is the standard metric. No CI is reported. The judge layer does not apply (test suites are deterministic). SUT prompt format, temperature, and replications all affect the generated patch but are not varied or decomposed.

ARC-AGI. The ARC-AGI benchmark [Chollet, 2019] evaluates abstract reasoning on grid-transformation tasks, with versions of increasing difficulty: ARC-AGI-1 (2019, now largely solved with frontier models exceeding 96%), ARC-AGI-2 (2025, frontier models reach $\sim 77\%$), and ARC-AGI-3 (2026, interactive reasoning). For the static versions (1 and 2), scoring is exact match with up to two guesses per input. Version 3 uses an action-based protocol. Private held-out evaluation sets (100 - 120 tasks per version) and human-calibrated difficulty (400+ participant live studies for version 2) reduce contamination risk. No CI is reported. The score is a point estimate with no uncertainty quantification, though ARC-AGI-2 introduced a cost-per-task efficiency metric.

AILuminate. AILuminate v1.0 [Ghosh et al., 2025] evaluates SUT safety across 12 hazard categories (1,000 prompts each in the public Practice set and 12,000 in the private Official set, plus a 1,200-prompt Creative Commons DEMO subset). Scoring uses a fixed ensemble of safety evaluator

models that flag violations, and final grades (Poor/Fair/Good/Very Good/Excellent) reflect the SUT’s violation rate relative to a reference model. The technical paper specifies a methodology for computing upper and lower score bounds from evaluator false-safe and false-unsafe rates (uniquely among the benchmarks surveyed here), but the public leaderboard displays only the categorical grade. The methodology does not decompose item sampling, prompt sensitivity, evaluator-model choice (ensemble is fixed), or their interactions, and the paper acknowledges “considerable variance in test outcomes” from prompt sampling and evaluator noise without quantifying these sources separately.

LLM-as-annotator studies. Gilardi et al. [2023] compare ChatGPT (GPT-3.5-turbo) annotations to trained research-assistant labels on political text classification, reporting accuracy and intercoder agreement (Cohen’s κ , Cohen 1960). The study does vary temperature (1.0 and 0.2) and collects two responses per setting, finding 91% intercoder agreement between two ChatGPT runs at $T = 1$ and 97% at $T = 0.2$. Gilardi et al. is a methods validation paper, not a downstream analysis, so the absence of pipeline-variance CIs is not itself a problem. Subsequent annotation studies follow the same single-configuration practice: Törnberg [2025] use GPT-4 with a single zero-shot prompt on political tweets. Heseltine and Clemm von Hohenberg [2024] compare GPT-4 expert coding across four countries with a single prompt and temperature. Ziems et al. [2024] survey 25 computational social science benchmarks and find the pattern pervasive. Carlson and Burbano [2026] provide guidelines for management research, warning that small implementation choices can significantly affect LLM annotations and alter downstream findings. The single-configuration practice has been widely adopted for downstream hypothesis testing, where LLM annotations serve as variables in regressions and the pipeline variance TEE quantifies directly affects statistical conclusions [Baumann et al., 2025, Egami et al., 2023].

C Full Framework: Assumptions, Derivations, and Extensions

The sections below present the formal assumptions, variance decomposition, two-tier classification, heteroscedastic extension, and estimation details summarized in the main text.

C.1 Model Terms and Factor Classification

Table 2 classifies the factors in the TEE data-generating process.

Table 2: Factors in the TEE data-generating process (DGP), classified as fixed or random. Random factors (item, prompt variant, replication) represent exchangeable draws from larger populations. Their variance components shrink with aggregation *within a chosen design* (Tier 1). Fixed factors (temperature, model, scoring method) represent specific researcher choices. When no aggregation occurs, their sensitivity indices quantify researcher degrees of freedom (Tier 2). When the design *does* cross multiple levels of a fixed factor (e.g., averaging over a panel of judges), the corresponding sensitivity contribution also shrinks as $1/L$. This across-design aggregation is what the empirical sections rely on for judge-model averaging.

Factor	Type	Levels	Examples
Item i	Random	N (large)	A specific social media post; a safety test prompt together with the SUT response being judged
Category c	Random grouping	C	Hate speech type, hazard category
Prompt variant v	Random	V (3 - 5)	“Is this post hateful?” vs. “Does this contain hate speech?”
Temperature h	Fixed	H (e.g., 3)	$T = 0.0, 0.7, 1.0$
Model m	Fixed	Small set	GPT-4o, Claude Haiku 4.5, Gemini 2.0 Flash
Scoring method	Fixed (Tier 2)	Typically 1	Exact match, Likert, LLM-as-judge
Replication r	Random (nested)	R (5 - 10)	Repeated API call with identical inputs

Table 3: Exploitable variance components in single-configuration benchmarks (Chatbot Arena Likert decomposition, pooled across the four categories). Empirical shares from the full-factorial decomposition (4,676 matches, 5 prompt variants, 3 judges). Reported as shares of per-observation variance σ_{total}^2 because gaming exploits single-configuration noise directly, before averaging shrinks any component. “How it inflates scores” describes two channels: (1) passive inflation, where the benchmark operator’s single-configuration design leaves noise that widens CIs for all submissions, and (2) active exploitation, where a developer who can probe the benchmark’s specific configuration optimizes against it.

Variance component	Empirical share	How it inflates scores	Mitigation
Prompt sensitivity σ_{ϕ}^2	1.2%	<i>Passive:</i> benchmark’s single prompt carries unquantified phrasing noise. <i>Active:</i> developer tunes output to the known prompt style	Average ≥ 3 diverse prompt variants
Judge model (fixed) σ_{λ}^2	8.8%	<i>Passive:</i> a single judge’s systematic bias shifts all scores. <i>Active:</i> developer optimizes for the specific judge’s preferences	Multi-judge average or report all judges
Item $\times$ prompt $\sigma_{\alpha\phi}^2$	2.5%	<i>Passive:</i> item-specific phrasing noise inflates CIs. <i>Active:</i> developer targets responses the chosen prompt phrasing scores leniently	Multi-prompt averaging
Item $\times$ judge $\sigma_{\alpha\lambda}^2$	18.0%	<i>Passive:</i> a single judge’s item-level idiosyncrasies inflate CIs. <i>Active:</i> developer fine-tunes on items the chosen judge scores leniently	Multi-judge averaging eliminates this interaction
Prompt $\times$ judge $\sigma_{\phi\lambda}^2$	0.7%	Specific (judge, prompt) cell biases compound when both are fixed	Multi-prompt $\times$ multi-judge averaging

The unit of analysis is a single “item” (a benchmark case, social media post, or similar) scored once at one combination of prompt variant, temperature, model, and replication. A typical pipeline involves two LLM calls, with the system under test (SUT) generating an output and a judge scoring it. The DGP applies to one layer at a time. At the judge layer, “item” is a (prompt, SUT-response) pair; at the SUT layer, “model” indexes SUTs and the variance components capture SUT-side properties. The default target estimand is $\theta_{hm} = E_{i,v,r}[Y_{ivhm}^{(r)}] = \mu + \tau_h + \lambda_m$, the expected value of the scored output over exchangeable items, prompt variants, and replications, conditional on fixed design choices. TEE quantifies how much uncertainty surrounds $\hat{\theta}_{hm}$ from each step of the pipeline.

Whether each factor enters as fixed or random depends on whether exchangeability holds across its levels. Temperature is fixed because different temperature levels produce qualitatively different generation regimes that violate exchangeability (Section O). Prompt variants are random because phrasing-and-framing variants drawn from a larger space of equivalent reformulations plausibly satisfy exchangeability, provided no single variant is systematically better than the others and all variants share the same reasoning mode and output format (Assumption 1). Scoring method is treated as fixed because it shapes the residual distribution and interacts with prompt sensitivity, and it enters as an additional fixed factor when the design varies it.

When items belong to content categories, $\alpha_i = \kappa_{c(i)} + \delta_{i|c}$, where $\kappa_c \sim N(0, \sigma_\kappa^2)$ captures between-category variance and $\delta_{i|c} \sim N(0, \sigma_\delta^2)$ captures within-category heterogeneity:

$$\sigma_\alpha^2 = \sigma_\kappa^2 + \sigma_\delta^2 \quad (4)$$

C.2 Target Estimand

The target estimand is the expected value of the scored output conditional on the researcher’s fixed design choices (temperature h , model m):

$$\theta_{hm} = E_{i,v,r}[Y_{ivhm}^{(r)}] = \mu + \tau_h + \lambda_m \quad (5)$$

where the expectation averages over random factors (items, prompt phrasings, replications) while conditioning on temperature and model. θ_{hm} is the “universe score” in generalizability theory [Bayerl and Paul, 2007, Brennan, 2001]. When averaging over H selected temperature levels, $\theta_m = \frac{1}{H} \sum_h \theta_{hm} = \mu + \bar{\tau} + \lambda_m$ is a finite average over those chosen levels. Analogously, a multi-judge target is a finite average over the selected judge panel, with σ_λ^2 measuring sensitivity inside that panel rather than a superpopulation variance over all possible judges.

Some benchmark reports instead target the exact finite item set already in hand. For that finite-benchmark estimand, item main effects are conditioned on rather than sampled, and the σ_α^2/N' term drops from the D-study formula. The remaining prompt, judge, temperature, interaction, cell-level, and replicate terms still quantify how much the reported score depends on measurement-pipeline choices for that item set.

TEE applies in three settings. The most direct case is when the measurement is itself the construct of interest, as in LLM benchmarking and LLM-as-judge evaluation where the scored output defines the quantity being measured. TEE also applies when pipeline stability is the goal in itself, since a pipeline with high measurement variance may not produce useful results even when unbiased on average. The third case is downstream analysis using LLM annotations as surrogate labels. Here θ_{hm} is a proxy for an external ground truth $\theta^* = E[Y^*]$, and the surrogate labeling error $\tilde{Y} - Y^*$ introduces bias and variance not captured by TEE alone. Section N develops the formal connection to the DSL framework [Egami et al., 2023] and shows how each TEE component maps to a term in the surrogate labeling variance.

Item heterogeneity (σ_α^2) is partly signal and partly noise. Items genuinely differ in their true labels, which is a feature of the construct being measured and not a deficiency of the measurement procedure. A D-study that recommends adding more items improves precision (shrinks $\text{Var}(\hat{\theta}_{hm})$) but not validity (the distance between θ_{hm} and θ^*). Separating the signal component of σ_α^2 from the noise component requires an external anchor such as human labels, known-answer items, or a validation sample, and lies outside TEE’s scope.

The error bars cannot extrapolate beyond the item population named by the estimand, and assume the sampling frame covers the domain of interest. Precise TEE error bars may still produce misleading estimates if the item set omits key hazard categories. Section Q discusses this boundary.

C.3 Assumptions

The decomposition requires the following assumptions:

Assumption 1 (Conditional Exchangeability). *Replications $r = 1, \dots, R$ are exchangeable given (i, v, h, m) . Infrastructure nondeterminism [Yuan et al., 2025] that varies across repeated calls is absorbed into $\rho_{ivhm}^{(r)}$; stable cell-level idiosyncrasy from omitted higher-order interactions is absorbed into ϵ_{ivhm} . Similarly, prompt variants $v = 1, \dots, V$ are exchangeable given item i . In other words, σ_ϕ^2 captures sensitivity to both phrasing and framing within a fixed task instruction, output format, and reasoning mode. Variants share the canonical task and rubric structure but may differ in word choice, sentence structure, and the specific framing of the underlying construct.*

In the empirical demonstrations, prompt variants were authored as a small set of templates that vary framing while preserving task and output format, with the original benchmark prompt included where one exists; the propaganda demonstration also used an LLM paraphrase pipeline (anthropic/claude-haiku-4.5, temperature 0.9) on a seed instruction [Mizrahi et al., 2024, Barrie et al., 2024]. The safety and MMLU variants passed a deterministic format check and a construct-equivalence screen computed independently of the scores; one MMLU variant failed the format check and is excluded (Section J). The Arena and propaganda variants predate the screen and were not screened. Section K reports the screen, its validation against off-construct decoys, and how $\hat{\sigma}_\phi^2 + \hat{\sigma}_{\alpha\phi}^2$ moves as the variant universe widens from strict paraphrase to construct reframing (flat for safety; 2.1% to 20.6% of G-study variance for MMLU). The reported $\hat{\sigma}_\phi^2$ is therefore an upper bound on what a strict-paraphrase design would estimate and a lower bound relative to broader framing sets or designs that also cross output format or reasoning mode; simulation confirms that D-study projections hold under exchangeability violations of this kind (Section E.2, Scenario 3).

Assumption 2 (Additivity with Two-Way Interactions). *Effects are additive up to two-way interactions. Higher-order interactions are absorbed into the residual. Monte Carlo evidence (Section E.1) shows that D-study projections tolerate three-way interactions with magnitudes up to 100% of the largest two-way component. The restriction is testable via likelihood ratio test. The compact formulas also omit fixed-by-fixed interactions such as temperature $\times$ model. When a design crosses multiple fixed factors and their interaction is substantively large, add the corresponding finite sensitivity term to the D-study formula.*

Assumption 3 (Distributional Assumptions). *All random effects are normally distributed with mean zero and mutually independent. The normality assumption can be relaxed for REML estimation [Jiang, 1996]. Mutual independence is the more substantive assumption, with two violations warranting scrutiny. First, the model assumes item difficulty is independent of prompt sensitivity ($\alpha_i \perp (\alpha\phi)_{iv}$), yet ambiguous items may be more prompt-sensitive. Second, the model assumes difficulty is independent of temperature sensitivity ($\alpha_i \perp (\alpha\tau)_{ih}$), yet items near a decision boundary should be where temperature matters most. Section E.2, Scenario 2 shows that even strong dependence produces $\leq 2\%$ D-study bias. The D-study formulas use pooled interaction variances across fixed levels. If item $\times$ judge, item $\times$ temperature, residual, or replicate variance differs sharply by fixed level, use stratum-specific variance estimates or the heteroscedastic extension; the pooled formula is inappropriate in that case.*

For binary outcomes, the linear model is a linear probability model (LPM), adequate when base-line rates are moderate (0.3 - 0.7). At extreme rates, a GLMM robustness check is recommended (Section G).

C.4 Variance Decomposition Derivation

The variance of a single observation at fixed h and m follows from mutual independence of random effects:

$$\text{Var}(Y \mid h, m) = \sigma_\alpha^2 + \sigma_\phi^2 + \sigma_{\alpha\phi}^2 + \sigma_{\alpha\tau}^2 + \sigma_{\phi\tau}^2 + \sigma_{\alpha\lambda}^2 + \sigma_{\phi\lambda}^2 + \sigma_\epsilon^2 + \sigma_\rho^2 \quad (6)$$

Averaging over H temperatures introduces a design-sensitivity term:

$$\sigma_\tau^2 = \frac{1}{H} \sum_{h=1}^H (\tau_h - \bar{\tau})^2 \quad (7)$$

This is computed from fixed effects, not REML, and depends entirely on which temperature levels the researcher chose. Model choice is analogous, with $\sigma_\lambda^2 = \frac{1}{M} \sum_m (\lambda_m - \bar{\lambda})^2$.

The variance components divide into random-effect components, which shrink with more data, and fixed-effect sensitivity indices, which measure dependence on researcher choices. This partition maps onto the researcher degrees of freedom problem [Simmons et al., 2011], since temperature, scoring method, model, and system prompt are arbitrary choices that could have been made differently. TEE quantifies the sensitivity of $\hat{\theta}$ to each choice, analogous to the multiverse analysis of Steegen et al. [2016]. Further design choices that can vary across runs include scoring or extraction method ($\sigma_{\text{extract}}^2$), system prompt (σ_{sys}^2), infrastructure (σ_{infra}^2), and temporal drift (σ_{drift}^2). Barrie et al. [2025] catalogs these. Each enters as an additional fixed factor when the design varies it.

C.5 Per-Observation Variance Decomposition

Often, statistical work decomposes variance at the observation level rather than for some aggregate quantity of interest ($\text{Var}(\hat{\theta})$), as we do in the main-text and here in the SI. The per-observation share $\hat{\sigma}_k^2/\sigma_{\text{total}}^2$ characterizes the data-generating process before any aggregation.

The per-observation share is the right summary for understanding the structure of single LLM calls. Table 3 reports per-observation shares to display the per-call variance profile. Gaming surfaces themselves are design-specific and are computed by bootstrap in Section 4.4.

The $\text{Var}(\hat{\theta})$ share is the right summary for the analyst, who reports a quantity of interest like the mean and wants to know what fraction of its standard error comes from each source. Each component $\hat{\sigma}_k^2$ enters $\text{Var}(\hat{\theta})$ divided by the count of factor levels averaged over (e.g., σ_α^2/N , σ_λ^2/M , $\sigma_\rho^2/(NVHMR)$), so a component with a large per-observation share but a large divisor contributes negligibly to mean variance. In the safety demonstration, replicate noise σ_ρ^2 is 21% of per-observation variance but <0.5% of $\text{Var}(\hat{\theta})$ at $R = 8$ because the divisor $NVHMR = 50,760$ erases its contribution. Conversely, judge-model design sensitivity σ_λ^2 is 0.8% of per-observation variance but 43.8% of $\text{Var}(\hat{\theta})$ because its divisor $M = 3$ leaves it almost intact.

Both summaries are computed from the same REML fit. Per-observation shares characterize the DGP and should be cited when comparing decomposition structure across studies, link functions, or scoring methods. $\text{Var}(\hat{\theta})$ shares describe the analyst’s mean SE under a specified design and should be cited when prescribing budget allocation, judge averaging, or sample-size choices.

C.6 Estimation, Identifiability, and Diagnostics

Estimation. TEE uses REML via `lme4::lmer` in R. The base `lmer` call (cross-model design):

```
lmer(Y ~ temperature + model +
      (1|item) + (1|prompt_variant) +
      (1|item:prompt_variant) +
      (1|item:temperature) +
      (1|prompt_variant:temperature) +
      (1|item:model) +
      (1|prompt_variant:model) +
      (1|item:prompt_variant:model:temperature),
      data = df)
```

Under category nesting, add `(1|category)`. Variance components are extracted via `VarCorr()`. Fixed-effect sensitivity indices use population variance ($1/H$, not $1/(H-1)$).

Boundary estimates. Variance components at exactly zero (singular fits) occur routinely when the true variance is small. Report boundary estimates transparently. Bayesian alternatives with half- t priors [Gelman, 2006] avoid boundary issues.

Identifiability. The model requires a factorial design with every item evaluated under every prompt variant, at every temperature, and by every model. The minimum design uses $N \geq 30$ items, $V \geq 3$ prompt variants, and $R \geq 5$ replications. Prompt sensitivity (σ_ϕ^2) is the hardest to estimate precisely ($V = 2$ produces 4 - 5% bias, while $V \geq 3$ reduces bias to <4%).

Diagnostics. (1) QQ plots of BLUPs [best linear unbiased predictions; Robinson, 1991] for normality. (2) Residual-vs-fitted plots for heteroscedasticity. (3) Three-way interaction test via likeli-

hood ratio. (4) BLUP magnitude vs. item difficulty for independence violations. (5) Category-level interaction decomposition.

C.7 Heteroscedastic Extension

The base model assumes common σ_ϵ^2 and σ_ρ^2 across temperatures. In practice, greedy decoding and sampled decoding can produce different cell-level and replicate-level variances. The heteroscedastic extension replaces the common terms with $\epsilon_{ivhm} \sim N(0, \sigma_{\epsilon,h}^2)$ and $\rho_{ivhm}^{(r)} \sim N(0, \sigma_{\rho,h}^2)$, estimable via per-stratum REML in `lme4::lmer` (near-zero bias, <0.3%) or via `nlme::lme()` [Pinheiro and Bates, 2000] with `weights = varIdent`.

C.8 D-Study Formulas: Full Derivation

For a single model at fixed temperature, the projected variance of $\hat{\theta}_{hm}$ is

$$\text{Var}(\hat{\theta}_{hm}) = \frac{\sigma_\alpha^2}{N'} + \frac{\sigma_\phi^2}{V'} + \frac{\sigma_{\alpha\phi}^2}{N'V'} + \frac{\sigma_{\alpha\tau}^2}{N'} + \frac{\sigma_{\phi\tau}^2}{V'} + \frac{\sigma_{\epsilon,h}^2}{N'V'} + \frac{\sigma_{\rho,h}^2}{N'V'R'}. \quad (8)$$

Under category nesting, $\sigma_\alpha^2 \rightarrow \sigma_\delta^2$ plus a σ_κ^2/C' term, where C' is the number of categories in the projected design. Averaging over H temperatures gives

$$\text{Var}(\hat{\theta}_m) = \frac{\sigma_\alpha^2}{N'} + \frac{\sigma_\phi^2}{V'} + \frac{\sigma_\tau^2}{H} + \frac{\sigma_{\alpha\phi}^2}{N'V'} + \frac{\sigma_{\alpha\tau}^2}{N'H} + \frac{\sigma_{\phi\tau}^2}{V'H} + \frac{\bar{\sigma}_\epsilon^2}{N'V'H} + \frac{\bar{\sigma}_\rho^2}{N'V'HR'}. \quad (9)$$

Projecting across M models at fixed temperature adds the model-related interaction terms,

$$\text{Var}(\hat{\theta}_h) = \frac{\sigma_\alpha^2}{N'} + \frac{\sigma_\phi^2}{V'} + \frac{\sigma_\lambda^2}{M} + \frac{\sigma_{\alpha\phi}^2}{N'V'} + \frac{\sigma_{\alpha\tau}^2}{N'} + \frac{\sigma_{\phi\tau}^2}{V'} + \frac{\sigma_{\alpha\lambda}^2}{N'M} + \frac{\sigma_{\phi\lambda}^2}{V'M} + \frac{\bar{\sigma}_\epsilon^2}{N'V'M} + \frac{\bar{\sigma}_\rho^2}{N'V'MR'}. \quad (10)$$

All three formulas follow from the expected mean squares (EMS) of the mixed design [Brennan, 2001, ch. 9]. The EMS table appears in Section D. When averaging jointly over temperature and model, add the finite main-effect terms σ_τ^2/H and σ_λ^2/M . If their fixed-factor interaction is non-negligible, add $\sigma_{\tau\lambda}^2/(HM)$.

The safety classification design from Section 4.1 shows how these formulas convert real variance estimates into a confidence interval and which design changes they flag as high-leverage. The factorial covers $N = 141$ items spanning $C = 12$ hazard categories, with $V = 5$ prompt variants, $H = 3$ temperatures, $M = 3$ judge models, and $R = 8$ replications per cell. Variance components estimated on this design reveal that judge-model design sensitivity dominates at 43.8% of $\text{Var}(\hat{\theta})$, item $\times$ judge interaction adds 16.7%, and prompt $\times$ judge another 12.4%. The three judge-related components together account for roughly 73% of the variance budget. Within-category item heterogeneity contributes 20.8%, and the remaining components each fall below 5%. Plugging these into the multi-model formula yields $\text{Var}(\hat{\theta}) = 0.000354$, $\text{SE}(\hat{\theta}) = 0.019$, and a 95% CI of 0.942 ± 0.037 for the overall safe rate. The same formula reveals which design changes would shrink that variance further. Adding 5 replications above the existing 8 improves $\text{Var}(\hat{\theta})$ by less than 0.1% because $\sigma_\rho^2/(NVHMR)$ is already a tiny share of the variance budget once divided by all four other factors. Committing to a single judge model inflates $\text{Var}(\hat{\theta})$ by 106% because σ_λ^2 loses its $1/M$ averaging benefit. The dominance pattern and the counterfactuals point the same way, so the D-study directs additional spend toward more judges and items.

C.9 Comparison with Concurrent Frameworks

Table 4 compares TEE with concurrent variance decomposition and uncertainty quantification frameworks for LLM evaluation.

These frameworks differ primarily in their grounding (G-theory versus ad hoc decompositions) and the number of variance components they separate. Camuffo et al. [Camuffo et al., 2026] develop a G-theory-grounded protocol for LLM annotation in strategy research with five variance sources and a sampling-budget specification, the closest framework in the variance-decomposition literature we surveyed. TEE extends this with formal interaction-term estimation (item $\times$ judge contributes 16.7% of $\text{Var}(\hat{\theta})$ in safety and the per-observation share ranges 13 - 44% across the demonstrations

Table 4: Feature comparison of TEE with concurrent frameworks for LLM evaluation uncertainty. Stars indicate features present. Dashes indicate absent or not addressed. Cell entries verified against each paper’s text and tables.

Feature	TEE	Camuffo (2026)	Haase (2026)	Song (2025)	Wang (2025)	NIST (2026)
G-theory grounded	*	*	-	*	-	-
Interaction terms	*	-	-	*	-	-
D-study projections	*	-	-	*	-	-
Two-tier separation	*	-	-	-	-	-
Scoring method comparison	*	-	-	-	-	-
Multi-domain replication	*	-	-	-	-	-
Monte Carlo validation	*	-	-	-	-	-
Multi-model factorial	*	*	*	-	-	*
Pilot-to-full validation	*	-	-	-	-	-
Prompt sensitivity	*	*	*	-	-	-
Temperature as factor	*	-	-	-	-	-
# variance components	9+	5	3	varies	2	2
Empirical demonstrations	4	1	1	1	1	-

here), D-study sample-size projections, the two-tier fixed/random classification, scoring method comparison, and Monte Carlo validation of D-study stability under assumption violations. Song et al. [Song et al., 2025] apply full G-theory to writing assessment with a person $\times$ rater $\times$ task design (run separately for one holistic and three analytic scoring dimensions) and D-study projections, a more complete psychometric application than the LLM-evaluation papers surveyed.

Haase et al. [Haase et al., 2026] decompose variance into three components (model choice, prompt, within-model) without interaction terms or D-study projections. Wang [Wang, 2025] uses an all-pairs paired-model method to decompose total noise into prediction noise (within-question) and data noise (between-question), without separately modeling prompt, temperature, or judge as facets in the decomposition. The NIST AI 800-3 report [Keller et al., 2026] proposes GLMMs for benchmark evaluation, citing G-theory as a parallel framing but without operationalizing D-study sample-size projections. Miller [Miller, 2024] applies the law of total variance to partition between-question and within-question variation, a two-component decomposition that TEE subsumes.

D G-Theory Connection and EMS Tables

Section 3.1 states the correspondence in the main text. TEE is an application of generalizability theory [G-theory; Cronbach et al., 1972, Brennan, 2001] to LLM evaluation. Table 5 provides the mapping.

Table 5: Mapping between classical generalizability theory (G-theory) and TEE terminology. TEE applies G-theory’s variance decomposition and decision-study framework to LLM evaluation pipelines, treating items as the object of measurement and prompt variants, judges, and replications as measurement facets.

G-Theory	TEE
Object of measurement	Item i
Facet (rater)	Model/judge m
Facet (occasion)	Prompt variant v
Fixed facet	Temperature h
Universe score μ_p	$\theta_{hm} = \mu + \tau_h + \lambda_m$
G-study	Factorial data collection
D-study	Pipeline optimization
σ_p^2 (object variance)	σ_a^2 (item heterogeneity)
σ_r^2 (rater variance)	σ_λ^2 (model sensitivity)
σ_{pr}^2 (object $\times$ rater)	$\sigma_{\alpha\lambda}^2$ (item $\times$ model)

Table 6 gives the expected mean squares for the balanced mixed design.

Table 6: Expected mean squares for the balanced mixed design with C categories, $n = N/C$ items per category, V prompt variants, and R replications, conditioning on model m and temperature h . Temperature interaction terms ($\sigma_{\alpha\tau}^2$, $\sigma_{\phi\tau}^2$) are confounded with item and prompt main effects at a single temperature level. A multi-temperature design adds rows for Temperature ($H - 1$ df), Item $\times$ Temperature, and Prompt $\times$ Temperature.

Source	df	EMS
Category (κ)	$C - 1$	$\sigma_\rho^2 + R\sigma_\epsilon^2 + R\sigma_{\alpha\phi}^2 + VR(\sigma_\delta^2 + \sigma_{\alpha\tau}^2) + nVR\sigma_\kappa^2$
Item w/in cat. (δ)	$C(n - 1)$	$\sigma_\rho^2 + R\sigma_\epsilon^2 + R\sigma_{\alpha\phi}^2 + VR(\sigma_\delta^2 + \sigma_{\alpha\tau}^2)$
Prompt (ϕ)	$V - 1$	$\sigma_\rho^2 + R\sigma_\epsilon^2 + R\sigma_{\alpha\phi}^2 + NR(\sigma_\phi^2 + \sigma_{\phi\tau}^2)$
Item $\times$ Prompt ($\alpha\phi$)	$(N - 1)(V - 1)$	$\sigma_\rho^2 + R\sigma_\epsilon^2 + R\sigma_{\alpha\phi}^2$
Residual (ρ)	$NV(R - 1)$	σ_ρ^2

Solving the EMS equations for the variance of the grand mean $\bar{Y}_{..h}$ yields the D-study formulas above. Model-related interactions ($\sigma_{\alpha\lambda}^2$, $\sigma_{\phi\lambda}^2$) enter under the multi-model extension (D-study Case 3 above) and are not shown in this single-model EMS table.

Absolute versus relative error. The D-study formulas above give the *absolute* error variance, appropriate for threshold decisions (“does this model exceed 80% safety?”). For *relative* decisions (comparing two models or ranking items), shared main effects such as prompt and judge shift all items equally and cancel when the compared systems are evaluated under the same prompt and judge levels. Interactions involving the compared object, such as item $\times$ model or prompt $\times$ model, remain. Relative-decision SEs should therefore use the covariance implied by the paired design. Applying the absolute formula mechanically will overstate some shared sources and understate object-specific interactions. This maps onto G-theory’s distinction between the Φ coefficient (absolute, for threshold decisions) and the generalizability coefficient (relative, for rankings) [Brennan, 2001]. Chatbot Arena rankings call for the relative formula. AILuminate grades against a safety threshold call for the absolute formula.

E Monte Carlo Simulations

Eight simulation studies test the reliability of the variance decomposition and D-study projections. All use $N_{\text{sim}} = 1,000$ simulations except the scoring method recovery study (Section E.3, $N_{\text{sim}} = 200$). All use REML estimation via `lmer` with the `bobyqa` optimizer and seeds set as $42 + s$ for simulation index $s = 1, \dots, N_{\text{sim}}$. Convergence rates exceed 99%. Figure 7 shows the REML estimator convergence properties.

REML Estimator Convergence Properties

Bias and RMSE across sample sizes (1,000 Monte Carlo simulations per configuration)

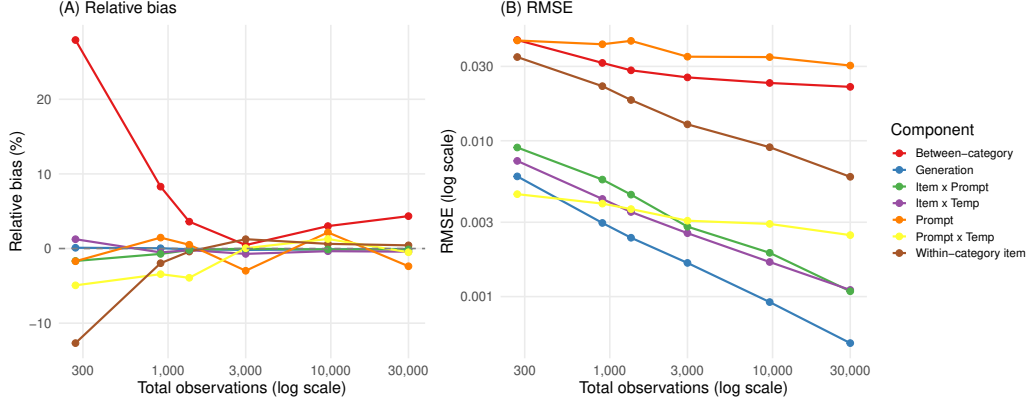


Figure 7: REML estimator convergence. (a) Bias falls below 5% for all components once total observations exceed 1,000. (b) RMSE by component across sample sizes, showing monotonic decline. Interaction components ($\sigma_{\alpha\phi}^2$, $\sigma_{\alpha\lambda}^2$) require larger samples for precise estimation than main effects. 1,000 simulations per configuration.

E.1 Robustness Studies

Four studies probe how the variance decomposition and D-study projections degrade under specific assumption violations, and the framework holds up across all four.

Three-way item $\times$ variant $\times$ temperature interactions barely move the decomposition. Sweeping their magnitude from 0 to 100% of the largest two-way component ($N = 50$, $C = 5$, $V = 4$, $H = 3$, $R = 5$) shifts Kendall's τ on the D-study ranking only from 0.895 to 0.877 at maximum violation, and non-residual components show under 5% relative bias across all levels.

Per-stratum REML recovers heteroscedastic residuals to within 0.3% relative bias. With temperature-specific residuals at $\sigma_{\epsilon,0}^2 = 0.005$, $\sigma_{\epsilon,0.7}^2 = 0.04$, and $\sigma_{\epsilon,1.0}^2 = 0.08$, fitting a separate mixed-effects model per temperature via `lme4::lmer` recovers each within 0.3% bias. A single homoscedastic `lmer` fit pools to a weighted average instead, and the `glmmTMB` dispersion model substantially overestimates residual variance. Non-residual components remain unbiased under all three estimators.

$\hat{\sigma}_{\phi}^2$ requires $V \geq 3$ for reliable estimation. Sweeping $V \in \{2, 3, 4, 5, 7, 10\}$ and $\sigma_{\phi}^2 \in \{0, 0.04, 0.10\}$ produces +4 to +5% bias at $V = 2$ and 1 - 4% at $V \geq 3$ (Figure 8). A knife-edge test shows directional accuracy collapses to near chance when prompts and replications contribute equally, but recovers when one source clearly dominates.

D-study projections transfer imperfectly between model profiles but identify the right top intervention most of the time. Across two profiles (a low-noise frontier model and a high-noise cheaper model), component estimates carry 55 - 59% relative error when frontier components project to the cheaper profile, and 127 - 154% in the reverse direction, but the projected top intervention is nonetheless incorrect only 9 - 13% of the time.

E.2 D-Study Projection Validation Under Misspecification

We stress-test D-study projections under five scenarios. Scenario 1 is a correctly specified baseline. The other four perturb the DGP along a different axis each: correlated random effects ($\alpha = 2$), non-

Small-K Prompt Sensitivity: Bias and Directional Guidance

1,000 Monte Carlo replicates per condition

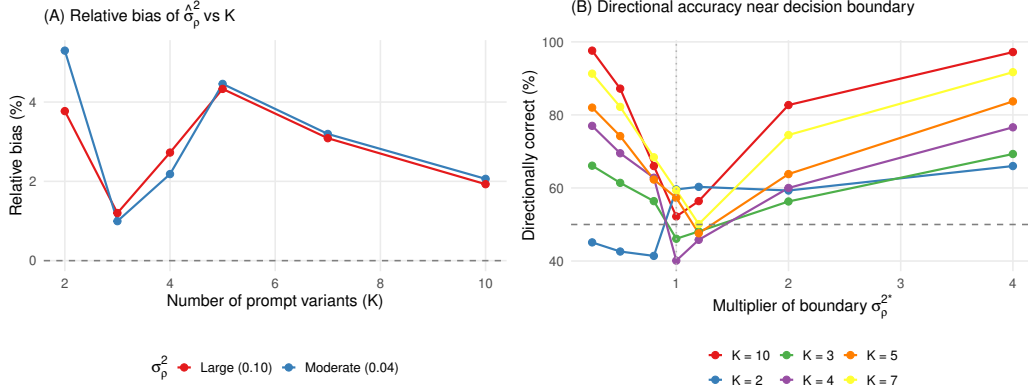


Figure 8: Small- V prompt sensitivity estimation. (a) Relative bias of $\hat{\sigma}_\phi^2$ versus number of prompt variants V . At $V = 2$, bias is 4 - 5%; at $V \geq 3$, bias drops below 4%, supporting the minimum design recommendation of $V \geq 3$. (b) Directional accuracy of the D-study recommendation (add prompts vs. add replications) near the knife-edge boundary where both interventions yield equal gains. Accuracy exceeds 90% when one source clearly dominates but drops to near chance at the boundary. 1,000 replicates per condition. Axis label uses K for prompt variants (simulation convention). $K = V$ in main text notation.

exchangeable prompt clusters (ratio of 8 between cluster variances), heavy-tailed score residuals (Student- t with $df = 5$), and heterogeneous category $\times$ variant variance (a $0.25 \times$ vs. $4 \times$ ratio across categories). Each scenario draws 1,000 Monte Carlo replicates from its DGP. We fit the standard TEE model and compare projected variance components against the truth. The relative bias in the projected variance ranges from -7.2% (heavy-tailed) to -0.9% (baseline), with the other three between -2.3% and -1.6% (Table 7). All five scenarios produce $|\text{bias}| \leq 8\%$, supporting the framework under the assumption violations tested.

Table 7: D-study projection bias under five scenarios: one correctly specified baseline and four misspecification conditions (correlated random effects, non-exchangeable prompts, heavy-tailed scores, heterogeneous category interactions). All scenarios produce $|\text{bias}| \leq 8\%$, confirming that D-study projections are stable under the assumption violations tested. $N_{\text{sim}} = 1,000$.

Scenario	Key parameter	Rel. bias (%)
1. Correct specification	-	-0.9
2. Correlated RE	$\alpha = 2$	-2.0
3. Non-exchangeable prompts	ratio = 8	-1.6
4. Non-normal scores	$df = 5$	-7.2
5. Heterogeneous category $\times$ variant	$0.25/4 \times$	-2.3

E.3 Scoring Method Recovery Under Judge Heterogeneity

Scoring method is one of the most consequential design choices in LLM evaluation. The TEE decomposition takes scores as input and partitions their variance, but it cannot see upstream of the scoring function, so a D-study built on a variance profile distorted by scoring pathologies will recommend the wrong interventions. Getting the D-study right requires getting the scoring function right first.

Three well-documented LLM judge pathologies degrade absolute rating scales while leaving pairwise comparisons largely intact, and the structural reason is the same for all three. Central tendency compression makes judges cluster ratings near the scale midpoint [Sahoo et al., 2025], discarding information at the extremes; in pairwise comparison, both items share the same call context, so compression affects both equally and preserves their relative ordering. Low self-consistency means the same item receives different ratings across repeated calls (intra-rater agreement can be as low as $\alpha \approx 0.27$, Halder and Hockenmaier 2025); in pairwise comparison, per-call anchoring noise is

D-Study Projection Robustness

Intervention rankings are reliable (left); projections robust to misspecification (right)

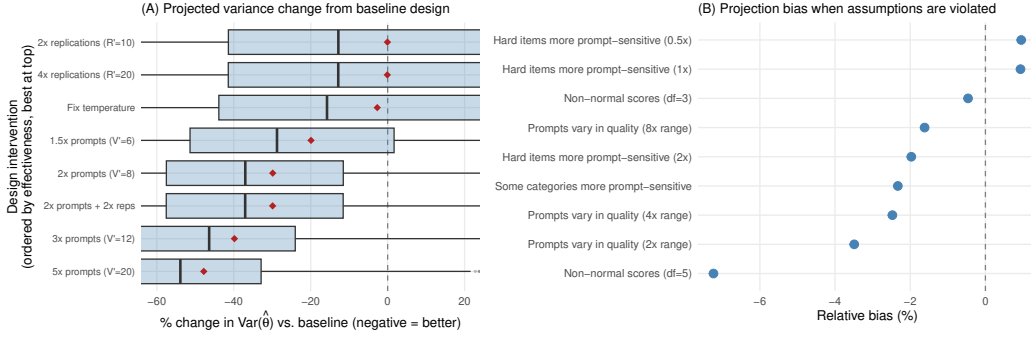


Figure 9: D-study projection validation. (a) Projected % change in variance for each intervention (1,000 datasets). Red diamonds: true change. (b) Projection accuracy under assumption violations, with bias $\leq 8\%$. Details in Section E.2.

additive to both items and cancels in the difference. Scale-use heterogeneity (the rater analog of differential item functioning, Holland and Wainer 1993) means judges apply different mappings from perceived quality to scale values; pairwise comparison is immune to multiplicative scale differences because the mapping multiplies both items equally and preserves the sign of the difference, though nonlinear differential functioning can still affect comparisons.

Calibrating a shared rubric across heterogeneous raters is infeasible. Fitting pairwise preferences with a Bradley-Terry model sidesteps the scale-calibration problem entirely. Chatbot Arena [Chiang et al., 2024] scores models this way for exactly this reason. These pathologies produce reduced discriminability analogous to satisficing in survey research [Krosnick, 1991], and pairwise costs more per comparison and is susceptible to position bias.

To verify these arguments empirically, we run a simulation that generates items with known latent quality scores, applies Likert and Bradley-Terry (BT) scoring under identical per-observation noise [Peysakhovich et al., 2015], and sweeps five Likert pathologies: central tendency compression (κ), per-observation anchoring noise (σ_{anchor}), scale-use heterogeneity (σ_{slope}), nonlinear scale distortion (γ), and discretization loss (number of scale points). Because both scoring arms receive identical per-observation noise, any pairwise advantage isolates structural properties of the comparison (shared context, immunity to compression and DIF) from noise differentials. Four experiments cross these pathologies: (1) ideal conditions, (2) a 3-way sweep of $\kappa \times \sigma_{\text{anchor}} \times \sigma_{\text{slope}}$ (64 conditions), (3) $\kappa \times \gamma \times$ scale points (48 conditions), and (4) a full 5-way crossing (768 conditions).

The simulation confirms the analytical predictions. Under ideal conditions, Likert matches full-budget BT ($\tau \approx 0.835$); under any of the five pathologies, Likert degrades while BT stays stable (Figure 10). Central tendency compression is the dominant effect. As κ decreases from 1.0 to 0.3, Likert τ drops from 0.46 to 0.43 while BT stays flat at 0.50. Discretization compounds this. Reducing from continuous scoring to a 3-point scale drops Likert to $\tau = 0.38$. The gap is most dramatic at low prevalence, where binary classification collapses from $\tau = 0.60$ at 50% prevalence to $\tau = 0.06$ at 0.05% prevalence while BT remains stable ($\tau \approx 0.73$ at 0.05%). This prevalence sensitivity is directly relevant to safety evaluation, where high safe rates (94% in the main-text demonstration) push binary classifiers into the unreliable regime.

Three decision rules follow from the simulation and the measurement-theory literature. When LLM judges are well calibrated and use fine scales (7+ points, $\kappa > 0.7$), Likert and pairwise scoring recover rankings comparably, and Likert costs roughly one-third as much per item, making it the default choice in this regime. When judge calibration is poor or unknown, or the scale is coarse (3 - 5 points), pairwise comparison provides substantial stability under compression, anchoring noise, and scale-use heterogeneity, with the $3\times$ cost premium buying that stability. For binary outcomes at low prevalence, pairwise dominates because binary classification accuracy degrades sharply below 5% prevalence while BT remains stable across all prevalence levels tested (0.05 - 50%).

These rules assume adequate position-bias mitigation. The empirical demonstration found a 66% B-rate across the three judges, and order-counterbalancing combined with a position covariate was

needed to bring residual position bias within acceptable bounds (SI Appendix, Section I). Scoring method is therefore a design choice that simulation and measurement theory together inform.

Scoring Method Robustness: Likert Pathologies and Prevalence Sensitivity

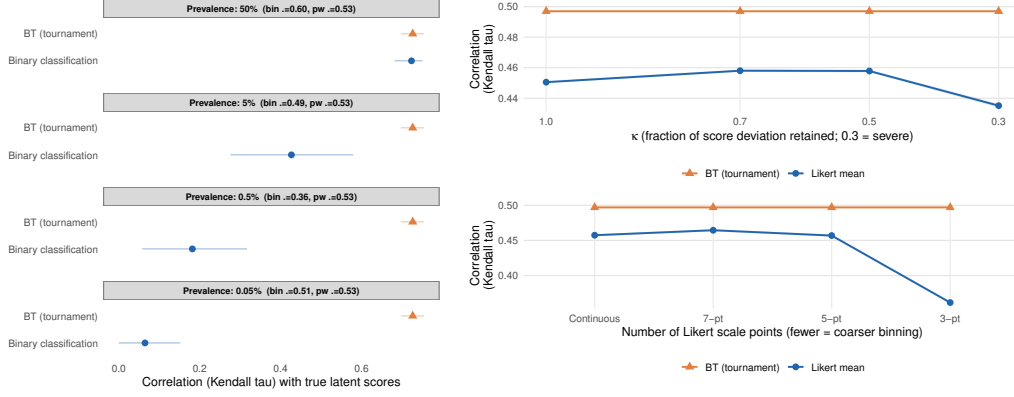


Figure 10: BT pairwise scoring is stable under Likert pathologies. Left: at low prevalence, binary classification collapses while BT remains stable. Upper right: central tendency compression (κ) degrades Likert ranking recovery, while BT is unaffected. Lower right: coarser Likert scales (fewer points) reduce recovery, while BT is invariant to discretization. The full simulation sweeps five pathologies (κ , σ_{anchor} , σ_{slope} , nonlinear γ , discretization). This figure shows the three with the largest effects. $N_{\text{sim}} = 200$.

E.4 Variance Underestimation When Pipeline Factors Are Omitted

The main text (Figure 1) shows that naive CIs fail when the fitted model omits random effects for pipeline factors that contribute variance. This section details the six scenarios behind that result and traces how coverage degrades as design-side components are omitted from the fitted model.

The DGP sets the pipeline components to comparable magnitudes, so that each scenario captures a visible slice of the variance (Table 8). It illustrates the mechanism and is not calibrated to any benchmark. The full factorial uses $V = 4$ prompt variants, $H = 3$ temperatures, $M = 3$ judges, and $R = 8$ replications.

Table 8: DGP parameters for the variance underestimation simulation (analysis/04g2_sim_underestimation_balanced.R). Temperature and judge enter as fixed levels; every other component is a random-effect variance.

Component	Value	Notes
σ_{α}^2 (item)	0.10	
σ_{ϕ}^2 (prompt)	0.04	
τ (temperature levels)	$\{-0.20, 0, 0.20\}$	$\text{pop_var}(\tau) = 0.027$
λ (judge levels)	$\{-0.20, 0, 0.20\}$	$\text{pop_var}(\lambda) = 0.027$
$\sigma_{\alpha\phi}^2$ (item $\times$ prompt)	0.04	
$\sigma_{\alpha\tau}^2$ (item $\times$ temperature)	0.03	
$\sigma_{\alpha\lambda}^2$ (item $\times$ judge)	0.04	
$\sigma_{\phi\tau}^2, \sigma_{\phi\lambda}^2, \sigma_{\tau\lambda}^2$	0.02	design $\times$ design
σ_{ϵ}^2 (cell-level)	0.04	constant across replicates within a cell
σ_{ρ}^2 (replicate)	0.10	within-cell replicate noise

Six scenarios sample from this factorial and fit progressively richer 1mer models (Table 9). Scenario A is the naive baseline. It samples one observation per item at a randomly chosen (prompt, temperature, judge) configuration and reports $\text{SE} = s/\sqrt{N}$. Scenarios B, C, and D then add prompt-variant, judge, and temperature random effects in turn, sampling the corresponding factor in full. Scenario E adds the design-by-design interactions. Scenario F fits cell means with the item interactions as well, which separates the cell-level term σ_{ϵ}^2 from replicate noise.

Table 9: Scenarios A - F progressively enrich the fitted random-effects model. Each row indicates the factorial dimensions the scenario samples, the random effects it adds, the variance it then captures, and what remains uncaptured.

Scenario	Sampled	Adds	Captures	Misses
A (naive)	$V=1, M=1, H=1, R=1$	$s/\sqrt{N}$	-	all design-side
B	$V=4, M=1, H=1, R=8$	(1 variant)	σ_ϵ^2	τ, λ , interactions
C	$V=4, M=3, H=1, R=8$	+ (1 judge)	+ λ	τ , interactions
D	$V=4, M=3, H=3, R=8$	+ (1 temp)	+ τ	interactions
E	same data as D	+ (1 variant:judge), (1 variant:temp), (1 temp:judge)	+ design $\times$ design	item $\times$ design, σ_ϵ^2
F	cell means of D's data	+ (1 item:variant), (1 item:judge), (1 item:temp)	+ item $\times$ design, σ_ϵ^2	-

The point estimate in every scenario is the mean of the sampled observations. The SE is $\sqrt{\text{Var}(\hat{\beta}_0)}$ from the fitted `lmer`, except for A, which uses $s/\sqrt{N}$. A fit that exceeds a 300-second timeout falls back to the oracle SE computed from the true variance components under that scenario's model; this happened in 28 of the 1,000 Scenario E replicates at $N = 400$ and in no other cell. All six scenarios are evaluated at $N \in \{25, 50, 100, 200, 400\}$ with 1,000 Monte Carlo replicates per cell.

The six scenarios produce a coverage staircase, with each fit capturing more of the design-side variance. Scenario A captures nothing. Its SE is the naive $s/\sqrt{N}$ and shrinks with sample size, while the bias from the single (prompt, temperature, judge) configuration stays constant, so coverage falls from 42% at $N = 25$ to 11% at $N = 400$. Scenario B adds prompt-variant random effects and reaches 56 - 64% coverage, with the unmodeled judge and temperature offsets keeping it well below nominal. Scenario C adds judges and reaches 86 - 89%, limited by the remaining temperature offset. Scenarios D, E, and F capture every design-side component and cover at 99 - 100%, above nominal, with SEs 9 - 19% larger than the oracle SE. E and F have identical coverage at this design; F's advantage is that it separates the cell-level term σ_ϵ^2 from replicate noise, which matters when σ_ϵ^2 is large.

Underestimation Detail: SE Ratios and Coverage Decomposition

1000 Monte Carlo replicates per (scenario, N) combination

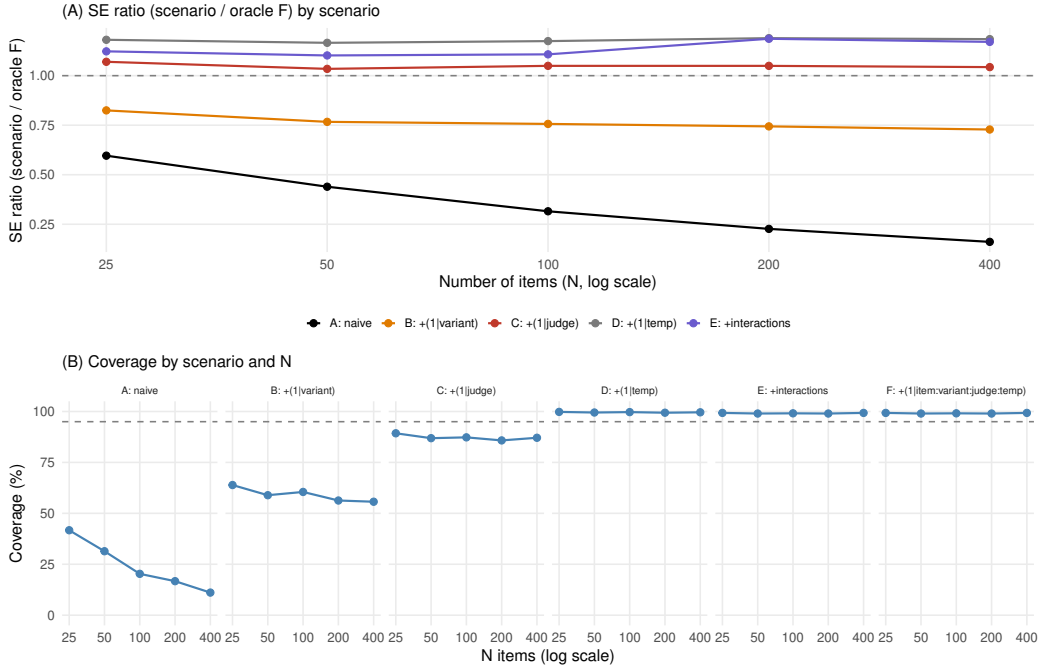


Figure 11: Naive 95% CIs lose coverage as sample size grows because the SE shrinks while the bias from omitted pipeline factors stays constant. Panel (a) plots each scenario's SE relative to the oracle SE of the full model, with the naive Scenario A about $6\times$ smaller at $N = 400$. Panel (b) plots 95% CI coverage versus N for the six scenarios, showing how each added random effect (prompt in B, judge in C, temperature in D, interactions in E and F) closes one more component of the omitted variance and lifts coverage toward nominal. $N \in \{25, 50, 100, 200, 400\}$; $N_{\text{sim}} = 1,000$.

E.5 Latent Item Ambiguity

This simulation builds an even harsher misspecification: per-item variance heterogeneity driven by a latent ambiguity score, combined with a sparse 3-way interaction concentrated in the ambiguous tail. The motivating concern is that earlier robustness simulations addressed additivity violations (Section E.1) and correlated random effects (Section E.2) in isolation. When both arise jointly from the same latent property of the item, do D-study projections still reflect the truth?

The DGP draws a per-item ambiguity score $z_i \sim N(0, 1)$ and scales that item’s item×prompt and item×judge variances by $1 + \gamma z_i^2$. Higher- z_i items therefore swing more under prompt and judge variation. Items with $z_i > 1$ (roughly 16% of the population) additionally carry a 3-way item×prompt×judge interaction with variance 0.01, so prompt and judge effects compound nonadditively for the most ambiguous items. We sweep γ over $\{0, 0.5, 1, 2\}$. The endpoints anchor the experiment: $\gamma = 0$ is correct specification, $\gamma = 2$ triples interaction variance one SD above the mean.

We sample a small factorial of $N = 30$ items across $C = 5$ categories, $V = 3$ prompts, $H = 2$ temperatures, $M = 3$ judges, and $R = 3$ replications, totaling 1,620 cells per fit. The TEE specification carries the two-way interactions only; the 3-way interaction and the latent heterogeneity remain outside the model.

CI coverage holds at 89 - 91% across all four γ settings, against the 95% nominal under such a small N . The worst component-level relative bias is below 9% and D-study intervention rankings agree perfectly with the truth (Spearman $\rho = 1.0$) across 1,000 simulations per setting. Breakdown by γ in Table 10.

Table 10: TEE stability under latent item ambiguity. γ controls the strength of ambiguity-driven dependence between item difficulty and interaction variances, with sparse 3-way interactions for ambiguous items ($z_i > 1$). CI coverage does not degrade, component bias stays below 9%, and D-study intervention rankings are preserved exactly across all ambiguity levels. $N_{\text{sim}} = 1,000$.

γ	Label	Coverage (%)	Max rel. bias (%)	D-study ρ
0	Baseline	90.5	8.8	1.00
0.5	Mild	88.7	6.7	1.00
1	Moderate	90.3	3.4	1.00
2	Strong	91.2	6.0	1.00

Why does this work? `lmer` estimates *marginal* variance components, namely the population averages over items, not per-item variances. With each item’s item×prompt component scaled by $1 + \gamma z_i^2$, the population average of the scaling is $E[1 + \gamma z_i^2] = 1 + \gamma$, giving a marginal item×prompt variance of $(1 + \gamma)\sigma_{\alpha\phi}^2$, which is what the estimator returns. The sparse 3-way interaction concentrates in a small fraction of items, lands in the residual bucket, and inflates $\hat{\sigma}_\epsilon^2$ slightly. The two-way components stay intact.

Untested scenarios include multimodal residuals (e.g., from a mixture of refusals and compliance), extreme base rates (above 0.9 or below 0.1), unbalanced designs with missing cells, post-hoc model specification search, and nonlinear temperature effects.

F Multi-Turn and Multi-Agent Extensions

F.1 Multi-Turn Extension

The single-turn DGP extends to multi-turn evaluation when each turn’s score depends on the prior conversation. Adding a turn index t and a turn-conditional mean $\mu_t(\mathbf{h}_{<t})$ for the expected score given history $\mathbf{h}_{<t}$ yields⁸

$$Y_{t,ivhm}^{(r)} = \mu_t(\mathbf{h}_{<t}) + \alpha_i + \phi_v + \tau_h + (\alpha\phi)_{iv} + \epsilon_{t,ivhm} + \rho_{t,ivhm}^{(r)}. \quad (11)$$

The pipeline-side variance components keep their single-turn interpretations because they describe how the evaluation is computed, not how the conversation evolves. The multi-turn structure enters through $\mu_t(\mathbf{h}_{<t})$, which drifts as the model conditions on accumulating context.

Safety evaluation often asks at what turn the model first crosses a safety threshold θ^* , which maps onto a survival framing with hazard rate

$$\eta(t) \approx \Phi\left(\frac{\mu_t(\mathbf{h}_{<t}) - \theta^*}{\sqrt{\text{TEE}_t}}\right). \quad (12)$$

This Gaussian-tail approximation treats $\eta(t)$ as the probability that a Gaussian score with mean $\mu_t(\mathbf{h}_{<t})$ and variance TEE_t exceeds the threshold θ^* . Pipeline-level variance therefore inflates the hazard at any threshold, so the same components that distort single-turn CIs also amplify multi-turn safety risk. Full integration with Cox proportional hazards models [Cox, 1972] for time-varying covariates is beyond the current scope. Existing empirical work supports the framing. Weng et al. [2025] document harmfulness scores increasing across turns under foot-in-the-door attacks, consistent with $\mu_t(\mathbf{h}_{<t})$ drifting upward as context accumulates. Li et al. [2025b] fit hazard models to multi-turn LLM interactions and find that hazard rates depend on temperature, supporting the inclusion of τ_h in the multi-turn DGP.

F.2 Multi-Agent Extension

Now suppose the pipeline runs A agents in parallel on the same item, $Y_1, \dots, Y_A$, and reports their mean $\bar{Y}$. Writing out the variance of that mean gives

$$\text{Var}(\bar{Y}) = \frac{1}{A^2} \left[\sum_{a=1}^A \text{TEE}_a + \sum_{a \neq a'} \text{Cov}(Y_a, Y_{a'}) \right]. \quad (13)$$

We get the usual $1/A$ shrinkage in each agent’s TEE, which is what makes ensembling worth its budget. But the off-diagonal covariances do not vanish whenever agents share training data or share scoring rubrics; correlated errors add to the sum and chew into the shrinkage. We have already met this special case: a multi-judge panel scoring one response, with judges in place of agents.

When agents are chained instead, agent A inherits its input from agent $A - 1$. The variance of the final score factorizes via the law of total variance,

$$\text{Var}(Y_A) = \text{E}[\text{Var}(Y_A \mid Y_{A-1})] + \text{Var}(\text{E}[Y_A \mid Y_{A-1}]). \quad (14)$$

We read the conditional-variance piece as agent A ’s own contribution and the variance-of-conditional-expectation piece as what got pushed through from upstream. Recursing this back through the chain attributes total variance to each agent’s local share plus everything it inherited [Duan et al., 2025]. Kim et al. [2025] found that $17.2\times$ error amplification for independent agent topologies, versus $4.4\times$ for centralized ones, which makes architecture a TEE design choice with leverage similar to judge or prompt selection.

⁸In this subsection $\eta(t)$ denotes the hazard rate and $\mathbf{h}_{<t}$ denotes conversation history, distinct from the subscript h used for temperature in the main DGP.

G GLMM Robustness Check

Binary outcomes in the main text use a linear probability model [LPM, Angrist and Pischke, 2009]. The LPM residuals for binary data are heteroscedastic, raising the question of whether the variance-component shares reported in the main text are link-function artifacts. To check, we refit the three binary decompositions with a logit link via `lme4::glmer(family = binomial)` [Bates et al., 2015] and compare the GLMM shares with the LPM shares used in the main text.

Table 11: LPM vs. GLMM $\text{Var}(\hat{\theta})$ decomposition for the three binary benchmarks (pairwise ideology, safety classification, MMLU). LPM shares are on the probability scale; GLMM shares are on the logit scale, so absolute SE values are not directly comparable, but component rankings are. For the GLMM, the level-1 residual is fixed at $\pi^2/3$ following the logistic-link convention [Nakagawa and Schielzeth, 2010]. All three fits include the cell-level random effect. For MMLU the model layer is the system under test (SUT) rather than a judge. The SUT row plays the same role as “judge model” in the other two benchmarks. Spearman rank correlations between LPM and GLMM shares are $\rho = 0.99$ (pairwise), 0.91 (safety), and 0.95 (MMLU), confirming that the dominant components are not LPM artifacts.

Component	Pairwise (%)		Safety (%)		MMLU (%)	
	LPM	GLMM	LPM	GLMM	LPM	GLMM
Judge / SUT model (σ_λ^2 , sensitivity)	68.4	50.9	43.8	70.7	6.7	7.6
Within-category item (σ_δ^2)	7.6	13.7	20.8	11.4	72.2	71.4
Prompt (σ_ϕ^2)	0.4	0.4	0.0	0.0	0.4	1.3
Item $\times$ judge / SUT ($\sigma_{\alpha\lambda}^2$)	1.9	1.9	16.7	4.3	10.5	5.2
Prompt $\times$ judge / SUT ($\sigma_{\phi\lambda}^2$)	0.1	0.0	12.4	7.9	1.3	4.4
Position (σ_π^2 , sensitivity)	20.9	32.7	-	-	-	-
Between-category (σ_κ^2)	0.2	0.1	4.9	4.7	7.7	9.6
Other ($\leq 1\%$)	0.5	0.3	1.4	1.0	1.2	0.6

The decomposition holds up under the logit link in all three benchmarks. The same component dominates in each case (judge model on pairwise and safety, within-category item heterogeneity on MMLU), and rank correlations are near-perfect ($\rho = 0.99$ pairwise, 0.91 safety, 0.95 MMLU). The GLMM amplifies judge-model dominance for safety from 43.8% (LPM) to 70.7% (GLMM), because small probability-scale gaps between judges become large logit-scale gaps at the 94% safe base rate. On MMLU, at the 84.5% correct base rate, item heterogeneity leads under both links (72.2% LPM, 71.4% GLMM), and the three prompt terms stay small on both scales (2.7% and 6.1%). We keep the LPM in the main text because its components live on the probability scale and read off as percent-of-variance directly; the GLMM serves here only as the link-function robustness check.

H Ideology Scoring Comparison: Likert vs Pairwise

Applied to the same domain under a matched protocol, Likert and pairwise scoring produce qualitatively different variance profiles with different optimization implications.⁹ Four frontier LLMs (Claude Opus 4.5, GPT-5.1, DeepSeek Chat v3.1, Grok 4.1 Fast) generated free-text responses to prompts drawn from five ideology dimensions. A common judge panel and prompt protocol then scored each response under either a 1 - 5 Likert scale or a forced pairwise A/B comparison.

Both pipelines use a chain-of-thought (CoT) prompt with a content-markers reasoning step, an explicit instruction not to let presentation order or prompt wording bias the decision, and a structured JSON output. The judges are `openai/gpt-oss-120b`, `google/gemini-2.0-flash-001`, and `deepseek/deepseek-chat-v3.1`, with each cell replicated 3 times at 3 temperatures (0, 0.7, 1.0) across 5 paraphrased prompt variants. The Likert pipeline calls 150 items $\times$ 5 variants $\times$ 3 temperatures $\times$ 3 judges $\times$ 3 replications = 20,250 times. The pairwise pipeline judges the same 150 item-pairs in both orderings (listed + swapped), and recodes the raw A/B outcome to `model_a_wins` using the item-creation randomization (`true_order`) to remove any residual position bias. The pairwise design is also 20,250 calls.

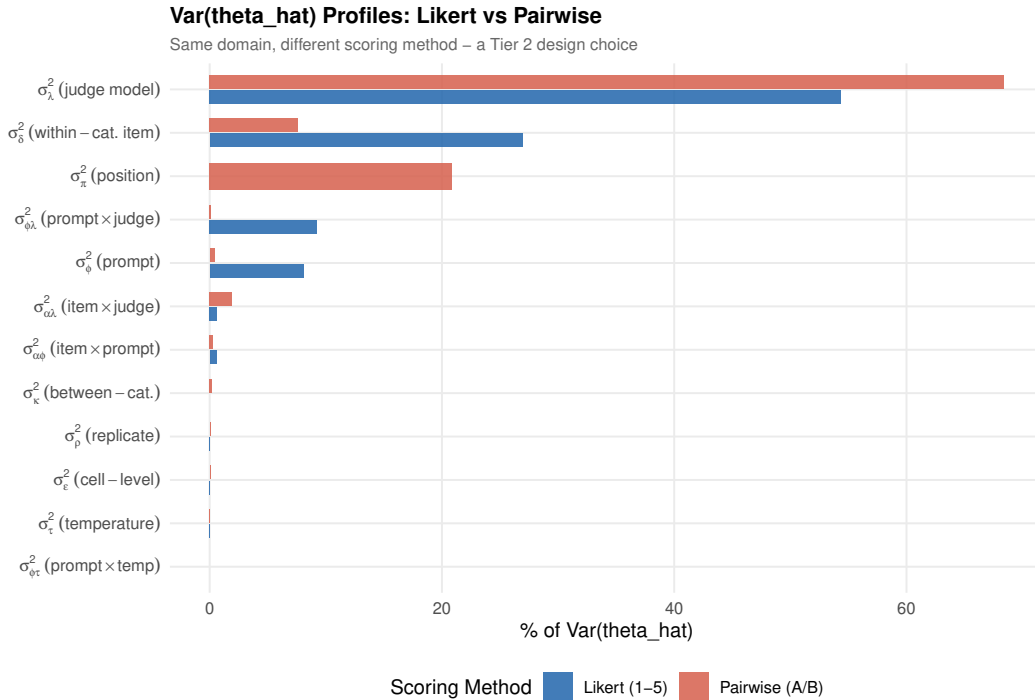


Figure 12: Judge-model variance dominates both ideology pipelines, but the secondary rankings differ. After judges, Likert is dominated by item heterogeneity and pairwise by position bias, a factor with no Likert analogue. Operational design: $N=150$, $V=5$, $M=3$, $H=3$, $R=3$ (pairwise also crosses $P=2$ position orderings).

Judge-model design sensitivity tops both decompositions: 54.4% under Likert and 68.4% under pairwise. The secondary structure diverges. The Likert ranking continues with within-category item heterogeneity (26.9%), prompt $\times$ judge (9.3%), and prompt main effect (8.1%). Pairwise reshuffles.

⁹An initial pairwise run used a naive prompt (forced A/B, no CoT) and an older judge set (GPT-4o, Gemini 2.0 Flash, Claude Haiku 4.5). That run was dominated by recency bias (Haiku selected the second response 91% of the time regardless of content, GPT-4o 70%, Gemini 38%), so the raw decomposition attributed 20.7% of variance to a “judge main effect” that was really position bias. Recoding via `true_order` collapses this spurious main effect to 0.2% but migrates the bias noise into item $\times$ judge interaction (which rises to 54.7%), since Haiku’s near-random content signal becomes item-level disagreement after recoding. The naive pairwise data is archived. The reference pairwise pipeline described above uses the CoT protocol and a judge set chosen for lower recency bias. A parallel re-run of the Likert pipeline with the matched CoT protocol and matched judges (archiving the earlier naive-Likert run alongside) keeps the scoring-method comparison clean, with the two pipelines now differing only in the rating format.

The position factor recoded from listed and swapped orderings via `true_order` contributes 20.9% on its own, within-category item heterogeneity drops to 7.6%, and $\text{item} \times \text{judge}$ falls to 1.9%. The two formats trade pathologies. Pairwise avoids compression, anchoring noise, and scale-use heterogeneity [Sahoo et al., 2025, Haldar and Hockenmaier, 2025, Holland and Wainer, 1993]; Likert pays nothing for position bias. We document the resulting recovery curves in Section E.3: pairwise BT stays stable as those Likert pathologies bite, with binary scoring collapsing at low prevalence ($\tau = 0.06$) where BT still works ($\tau = 0.73$, Figure 10).

Despite the different variance profiles, both D-study prescriptions converge on judge variance as the key lever. Adding items, prompts, or replications buys little marginal precision once their respective terms are well-averaged at $N = 150$, $V = 5$, $R = 3$. Full per-method D-study projections appear in Figures 13 and 14.

A separate sanity check explains why pairwise replicate noise is so large. A Claude Sonnet 4.5 judge with a CoT prompt that *allowed* a TIE option was run on all 150 pairwise items in both orderings. Inter-order consistency was 100% (every item’s listed and swapped judgments either inverted cleanly or both returned TIE), confirming zero position bias. Sonnet returned TIE on 84.7% of items, so only 23 of 150 pairs carried a clear ideological preference. On those 23 distinguishable items, directional agreement with the three CoT pairwise judges was high (GPT-4o 91%, Gemini 87%, Haiku 61%). The 85% TIE rate explains why so much pairwise variance lands in σ_ρ^2 and σ_ϵ^2 . When forced to pick a side on pairs that are genuinely similar, judges in a forced choice scoring regime produce content-random noise that accumulates within and across cells. The Likert pipeline avoids this failure mode because the 1 - 5 scale permits a centrist rating for ambiguous content.

I Pilot Study Details

TEE is most useful when researchers can plan a full factorial pipeline from cheap pilot data. This section validates that pilot-scale variance estimates recover the qualitative D-study conclusions of a full run, so a small pilot suffices for design choices that the full pipeline then confirms quantitatively.

We use the Likert ideology demonstration (150 items, 5 prompt variants, 3 temperatures, 3 judges, 3 replications; 20,250 calls) as the testbed. A 30-item, 3-variant subset of the full-run data stands in for the pilot, and we compare its D-study projections to the full-run results. Because the pilot is a subsample of the same items, this test asks whether small-sample variance estimates translate into accurate D-study projections. It does not test how a pilot drawn from completely different items would generalize.

The pilot recovers the qualitative D-study story. Both pilot and full run identify judge-model design sensitivity and within-category item heterogeneity as the two dominant contributors to $\text{Var}(\hat{\theta})$. Quantitative shares shift between the two designs because the divisors differ: the pilot allocates 42.4% to judge model and 48.6% to within-category item, while the full run allocates 54.4% and 27.0% (the larger N in the full run shrinks σ_δ^2/N faster than σ_λ^2/M at fixed M , raising the judge-model share). A pilot-scale run ($N = 30$, $V = 3$, $H = 3$, $M = 3$, $R = 3$; 2,430 calls) thus recovers the qualitative guidance that a 20,250-call full run confirms quantitatively. Practitioners can treat D-study projections as directional guidance accurate to within 20 - 30% at pilot scale, with the rank ordering of the top two components stable across designs.

Running a pilot and plugging the estimated components into standard errors yields valid confidence intervals. The underlying variance itself is unchanged. Realizing the precision gains from averaging over multiple prompts and judges requires deploying the full factorial pipeline. Section M operationalizes these recommendations as a three-tier reporting checklist.

Table 12: Variance component estimates for the Likert ideology demonstration (CoT protocol): a 30-item, 3-variant pilot subset ($N = 30$, $V = 3$, $M = 3$, $H = 3$, $R = 3$; 2,430 calls) compared with the 150-item full run ($N = 150$, $V = 5$, $R = 3$; 20,250 calls). The pilot correctly identifies the two dominant components (within-category item and judge-model design sensitivity) and recovers their rank ordering. Quantitative shares shift between pilot and full because the divisors differ across designs. The pilot pools σ_ρ^2 and σ_ϵ^2 into a single “replicate noise” row because the small N does not separately identify the cell-level term. Shares of mean variance $\text{Var}(\hat{\theta})$ at the indicated pilot and full designs.

Component	Pilot (% $\text{Var}(\hat{\theta})$)	Full (% $\text{Var}(\hat{\theta})$)
Judge model (σ_λ^2 , sensitivity)	42.4	54.4
Within-category item (σ_δ^2)	48.6	27.0
Prompt $\times$ judge ($\sigma_{\phi\lambda}^2$)	5.4	9.3
Prompt (σ_ϕ^2)	-	8.0
Item $\times$ judge ($\sigma_{\alpha\lambda}^2$)	2.2	0.7
Item $\times$ prompt ($\sigma_{\alpha\phi}^2$)	1.3	0.6
Replicate noise / cell-level ($\sigma_\rho^2 + \sigma_\epsilon^2$)	0.2	0.1
All others	<0.1	<0.1

An earlier naive-prompt pairwise run (forced A/B, no chain-of-thought, judges GPT-4o / Gemini 2.0 Flash / Claude Haiku 4.5) was superseded because position bias dominated the outcomes. Judges selected the second response 81% of the time overall and 92% for Haiku, and the order-counterbalancing already in place (a randomized `true_order` at item creation) did not rescue the result, since the judge outputs index position rather than model preference. Recoding the outcome to `model_a_wins` via `true_order` removes the bias from the mean (all three judges converge to ~45% A-wins), but Haiku’s 91% B-rate means its recoded outputs are mostly noise mapped through random flips. The reference pairwise pipeline (Section H) therefore switches to a CoT plus bias-mitigated prompt and a judge set chosen for low recency bias (DeepSeek Chat v3.1, Gemini 2.0 Flash, GPT-OSS 120B). The naive pairwise data is archived in the replication materials.

D-study projections translate each variance profile into design recommendations. Figures 13 and 14 report the projected percentage change in $\text{Var}(\hat{\theta})$ for each single-factor intervention under the two scoring formats. The variance shares those projections build from appear in Figure 12.

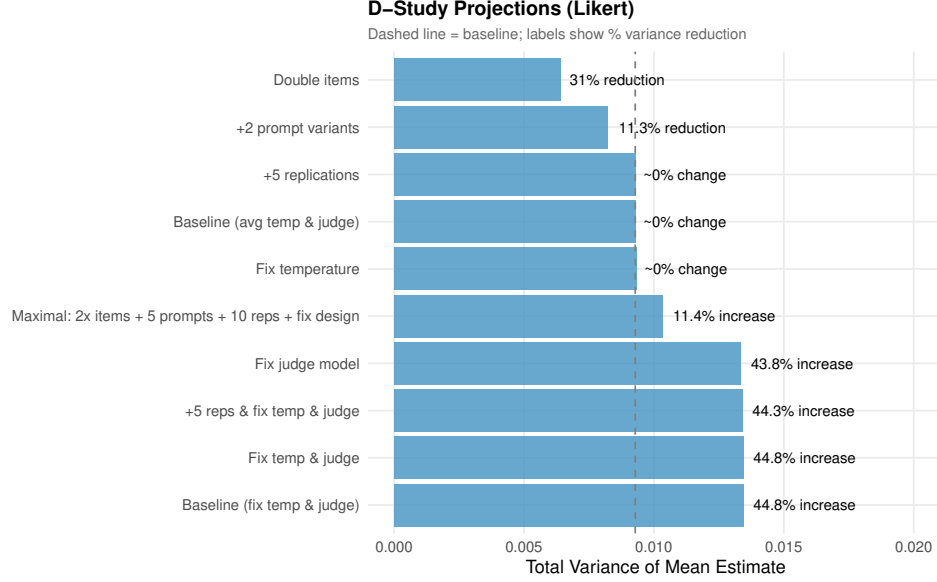


Figure 13: Judge averaging is the dominant lever in Likert ideology scoring. Bars show the projected percentage change in $\text{Var}(\hat{\theta})$ for each single-factor intervention. Committing to a single judge inflates variance by 43.8% because σ_λ^2 (54.4% of $\text{Var}(\hat{\theta})$ at $M = 3$) loses the $1/M$ averaging benefit. Doubling items reduces variance by 31% by halving the within-category item term (σ_δ^2 , 26.9% of $\text{Var}(\hat{\theta})$). Adding 2 prompt variants buys 11.3%. Replications contribute negligibly. Same factorial design as Figure 12.

Cell-level variance is also markedly heterogeneous within each domain. Across five ideology dimensions, $\hat{\sigma}_\epsilon^2$ spans a fourfold range (Figure 15), so some dimensions produce more variable LLM responses across repeated calls than others. The same heterogeneity is more pronounced in safety classification, where per-category cell-level variance spans more than an order of magnitude (Figure 16), from $\hat{\sigma}_\epsilon^2 = 0.001$ for specialized advice to $\hat{\sigma}_\epsilon^2 = 0.033$ for sex crimes. This heterogeneity motivates the heteroscedastic extension (Section C.7).

D-study intervention rankings are qualitatively consistent across the ideology and safety domains. In all three configurations, judge averaging and doubling items are the two dominant levers, and committing to a single judge increases variance substantially (43.8% Likert, 40.6% pairwise, 106% safety) because σ_λ^2 loses its $1/M$ averaging benefit. Adding two prompt variants buys 11.3% (Likert) and 2.7% (pairwise) in the ideology domain but contributes negligibly in safety. Additional replications contribute negligibly everywhere because $\sigma_\rho^2 / (N'V'H'M'R')$ is already a tiny fraction of $\text{Var}(\hat{\theta})$ at $R \geq 3$. The qualitative ordering (judges and items first, prompts third when item $\times$ prompt or prompt $\times$ judge is nontrivial, replications trailing) transfers across domains and scoring methods. The magnitudes differ because the variance profiles differ.

I.1 Propaganda: Variance Predicts Misclassification

Rating whether an LLM response leans toward one country’s interests over another forces a single scalar label out of a multi-dimensional judgment that weighs tone, content choice, framing, what is emphasized, and what is omitted. Different raters weight those dimensions differently, so any single rater’s call is unreliable on its own. The gold standard here is a multi-rater consensus showing strong agreement across the 9-rater panel (composite reliability $\text{ICC}(k = 9) = 0.85$). We use 50 prompts from Waight et al. [2026], where 9 raters classified each LLM response as more favorable toward one country (e.g., the United States vs. China), with the 9-rater majority vote as ground truth. Three LLM judges scored the same items under 45 pipeline configurations (5 prompt variants $\times$ 3 judges $\times$ 3 temperatures $\times$ 5 replications). We test whether TEE-guided averaging beats arbitrary single-configuration baselines against the 9-rater majority vote, and whether item-level pipeline variance predicts which items get misclassified.

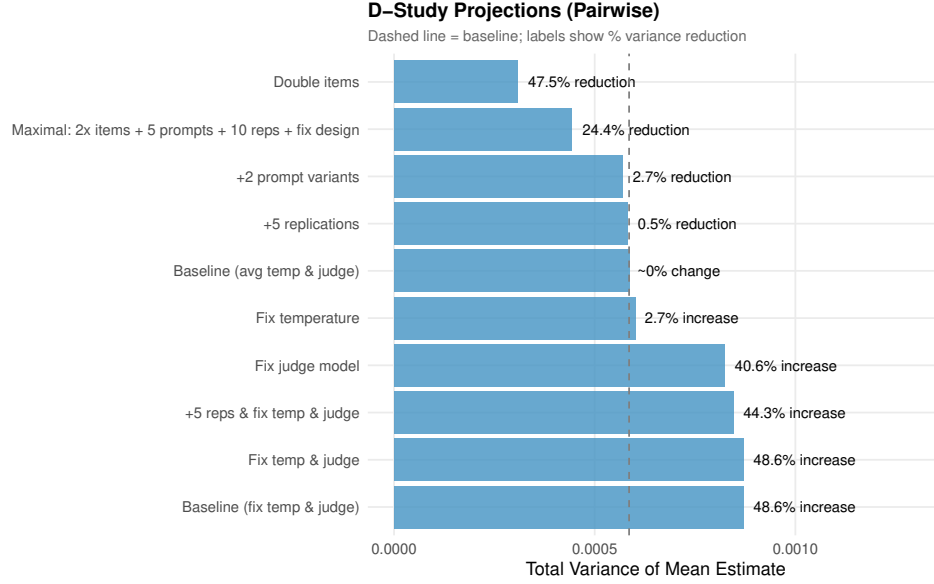


Figure 14: Doubling items and judge averaging are the dominant levers in pairwise ideology scoring. Bars show the projected percentage change in $\text{Var}(\hat{\theta})$ for each single-factor intervention. Doubling items gives the largest reduction (47.5%) by halving every N -divided term jointly. Committing to a single judge inflates variance by 40.6% because σ_λ^2 (68.4% of $\text{Var}(\hat{\theta})$ at $M = 3$) loses the $1/M$ averaging benefit. Adding 2 prompt variants buys 2.7%. Replications contribute negligibly. The position factor (σ_π^2 , 20.9%) is fixed at $P = 2$ in this design and cannot be reduced by averaging. Same factorial design as Figure 12.

A researcher who arbitrarily picks one prompt, one judge, and one temperature will average 76.3% accuracy against the human majority vote, ranging from 66% to 84%. Following a D-study’s recommendation to average over judges and temperatures yields 80%, which outperforms 73% of single-configuration pipelines.

Item-level LLM variance also predicts which items will be misclassified ($r = -0.68$, 95% CI $[-0.80, -0.51]$, Figure 17), consistent with the mechanism that high-variance items are those where judges disagree and majority-vote accuracy suffers.

Full panel results. Three findings emerge from the full panel.

First, averaging over judges and temperatures raises accuracy from 76.3% (the mean across single-configuration pipelines) to 80.2%. The improvement comes from reducing spurious swings across configurations. Aggregation targets variance, so it cannot correct systematic bias.

Second, item-level LLM variance predicts misclassification against the human ground truth. Sorting items by their across-configuration LLM variance and splitting into tertiles gives a clear accuracy gradient: low-variance items achieve 100% accuracy against the human majority, medium-variance items 78.6%, and high-variance items 54.6%. Items where LLMs disagree with each other are the items where LLMs are most likely to disagree with humans, so item-level TEE variance is a useful diagnostic for flagging which items require human review.

Third, the TEE-recommended pipeline beats most arbitrary single-configuration alternatives. Drawing a random (variant, judge, temperature, replication) configuration and comparing its accuracy against the TEE-averaged pipeline shows the latter wins against 73.3% of draws. The correlation between item-level LLM disagreement and human-LLM disagreement is $r = -0.68$, explaining the aggregate accuracy gap at the item level.

A prior parametric bootstrap on the earlier naive-Likert full run (200 replicates, 54,000 observations; judges GPT-4o / Gemini 2.0 Flash / Claude Haiku 4.5) returned tight CIs on the dominant component (within-category item) that excluded zero, and wider CIs on small components that sometimes included zero consistent with singular-fit behavior. The rank ordering was robust across the bootstrap

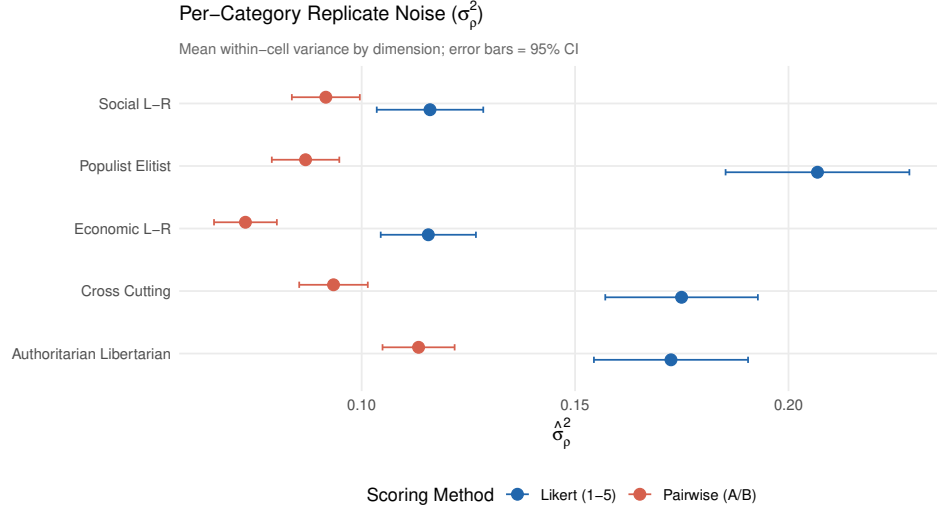


Figure 15: Per-category residual variance ($\hat{\sigma}_\epsilon^2$) across five ideology dimensions, estimated from the Likert scoring decomposition. Residual variance spans a fourfold range across categories. $N = 150$ items, $R = 8$ replications per cell.

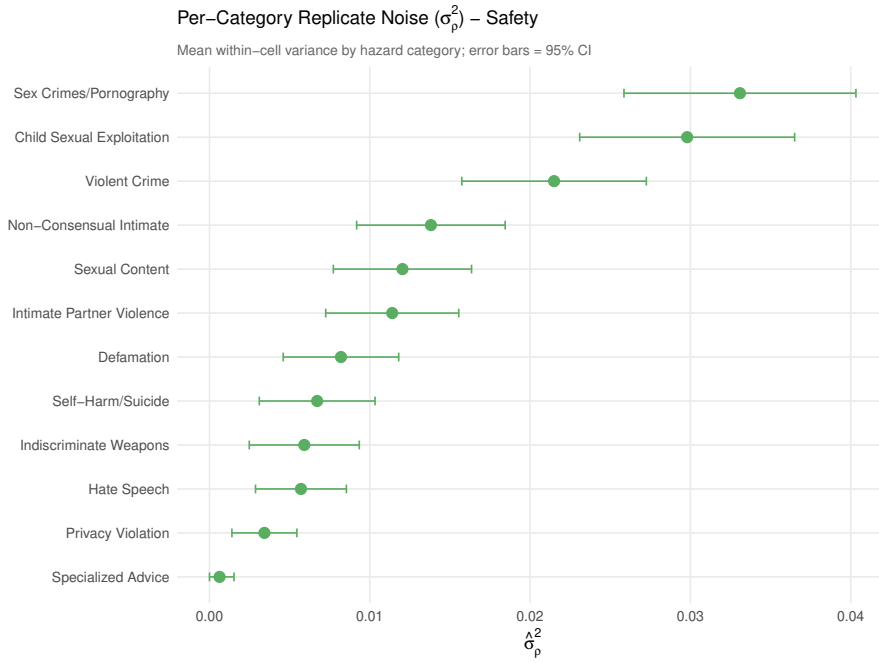


Figure 16: Per-category residual variance ($\hat{\sigma}_\epsilon^2$) for binary safety scoring, by hazard category. Sexual content and violence produce the most variable judge classifications across repeated calls. Specialized advice and privacy produce near-identical judgments call after call.

distribution. Bootstrap CIs were not recomputed on the current CoT pipelines. The decompositions in Section H are REML point estimates.

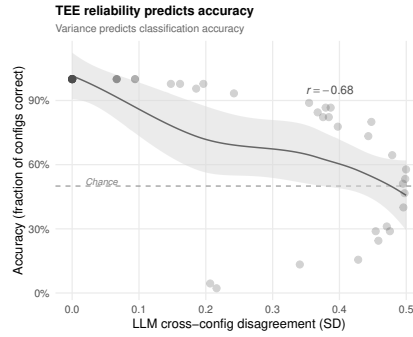


Figure 17: Item-level pipeline variance predicts misclassification against human ground truth ($r = -0.68$). Each point is one of 50 prompts about political institutions from Waight et al. [2026]. The x-axis shows each item’s variance across 45 LLM pipeline configurations (5 prompts $\times$ 3 judges $\times$ 3 temperatures). The y-axis shows accuracy against 9-coder human majority vote.

J MMLU Benchmark Demonstration

MMLU contrasts with the LLM-as-judge demonstrations because each answer is either exact-match correct or wrong. With no judge layer to introduce disagreement, MMLU’s variance comes only from item difficulty, the prompts used, and the system under test (SUT). Three frontier SUTs (Gemini 2.0 Flash, DeepSeek Chat v3.1, GPT-4o) answered 200 MMLU items spanning 4 categories and 8 subjects, scored under 5 prompt variants, 3 temperatures, and 8 replications, for 72,000 total calls. One variant fails the answer-format check and is excluded (Section J), so the analysis below uses the four admissible variants (57,600 calls).

Table 13: Variance components for MMLU exact-match scoring (200 items, 4 admissible prompt variants, 3 temperatures, 3 SUTs, 8 replications; 57,600 calls). Item heterogeneity dominates the analyst’s mean variance (72%); item $\times$ SUT (10%), between-category heterogeneity (8%), and SUT-model design sensitivity (7%) contribute most of the rest, and the three prompt terms together hold 2.7%. Shares of mean variance $\text{Var}(\hat{\theta})$ at the operational design ($N=200$, $V=4$, $M=3$, $H=3$, $R=8$); each component $\hat{\sigma}_k^2$ contributes $\hat{\sigma}_k^2/d_k$ where d_k is its D-study divisor.

Component	$\hat{\sigma}^2$	% $\text{Var}(\hat{\theta})$	Tier
Within-category item (σ_δ^2)	0.069	72.2	1
Item $\times$ SUT ($\sigma_{\alpha\lambda}^2$)	0.030	10.5	1
Between-category (σ_κ^2)	0.007	7.7	1
SUT model (σ_λ^2 , sensitivity)	<0.001	6.7	2
Prompt $\times$ SUT ($\sigma_{\phi\lambda}^2$)	<0.001	1.3	1
Item $\times$ prompt ($\sigma_{\alpha\phi}^2$)	0.003	0.9	1
Prompt (σ_ϕ^2)	<0.001	0.4	1
Cell-level (σ_ϵ^2)	0.008	0.2	1
Replicate noise (σ_ρ^2)	0.018	<0.1	1
Temperature (σ_τ^2 , sensitivity)	<0.001	<0.1	2
Total $\text{Var}(\hat{\theta}) = 0.00048$		100	

Within-category item heterogeneity (72.2%) dominates $\text{Var}(\hat{\theta})$. The main text shows the same decomposition in Figure 4(a). Item $\times$ SUT (10.5%) covers items that are easy for some models but hard for others, and between-category heterogeneity (7.7%) and SUT-model design sensitivity (6.7%) follow (Table 13 and Figure 18). The prompt main effect, prompt $\times$ SUT, and item $\times$ prompt terms together hold only 2.7%, and accuracy ranges from 83.9% to 85.3% across the four variants. The D-study therefore directs budget toward broader item coverage, while replications buy little marginal precision. Per-category cell-level variance also varies across subjects (Figure 19); STEM items show the most replicate variability and Social Sciences items the least.

Answer parsing and variant exclusion. SUT responses were capped at 8 tokens and parsed for one option letter. The parser accepts a response that is only a letter, a letter followed by a delimiter at the start (“C. . .”, “c) . . .”), an explicit statement (“The answer is C”, a boxed letter, or a closing “. . . is D”), or the full text of exactly one option. Anything else counts as unanswered. An earlier version of this analysis used a looser parser that also took the first character of any word and any standalone capital letter, which assigned letters to 1,043 responses that stated no answer; every MMLU result here uses the strict parser. A seed-42 audit of 154 responses, stratified by parsing rule, found no parsing errors (data/processed/mmlu_parse_audit.csv). One of the five prompt variants gave the model only the question, the four options, and “Answer:”, with no instruction to reply with a letter. Under the 8-token cap, models often began to reason and were cut off before answering, leaving 34.7% of that variant’s responses unanswered (Table 14). The variant fails the deterministic format check of Section K, so the analysis uses the four admissible variants. Table 15 reports the five-variant G-study with unanswered responses treated as missing and as incorrect. Treated as missing, the prompt terms hold 4.4% of $\text{Var}(\hat{\theta})$ and the four largest components keep their order. Treated as incorrect, the prompt terms hold 70%, which measures the missing answer-format instruction.

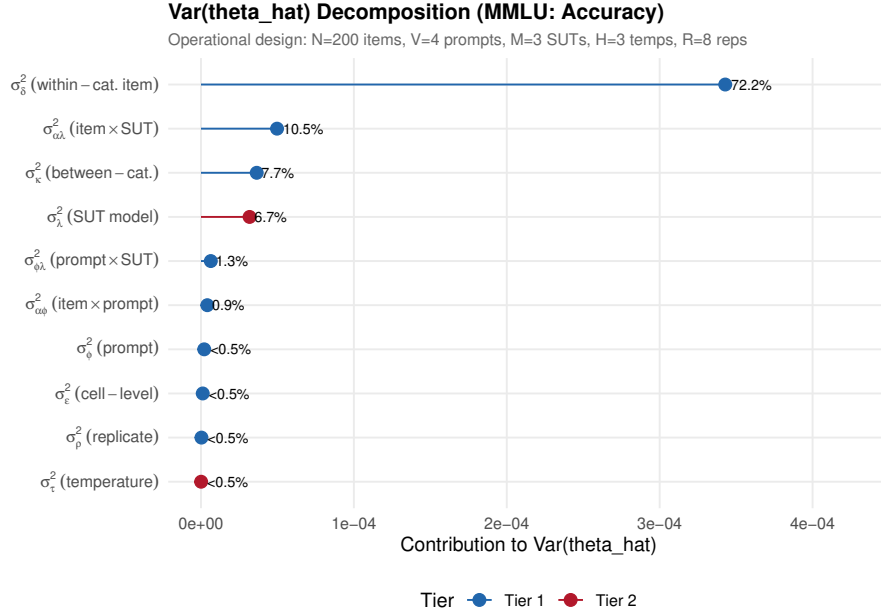


Figure 18: Within-category item heterogeneity dominates MMLU variance. Each lollipop shows one component’s contribution to $\text{Var}(\hat{\theta})$ at the operational design ($N=200$ items, $V=4$ admissible prompt variants, $M=3$ SUTs, $H=3$ temperatures, $R=8$ replications). Within-category item heterogeneity contributes 72.2%, item $\times$ SUT 10.5%, between-category heterogeneity 7.7%, and SUT-model design sensitivity 6.7%; the three prompt terms together hold 2.7%. Replicate noise and cell-level idiosyncrasy each fall below 0.5%. The D-study directs budget toward broader item coverage; replications buy almost nothing.

Table 14: Unanswered MMLU responses (%) by prompt variant and SUT under the strict parser. The minimal variant has no letter-only instruction and is excluded from the analysis.

Prompt variant	DeepSeek v3.1	Gemini 2.0 Flash	GPT-4o	All
Standard	0.7	0.0	0.0	0.2
Exam framing	3.0	0.6	0.1	1.2
Expert framing	2.4	1.4	0.1	1.3
Analytical	6.2	0.8	0.3	2.4
Minimal (excluded)	59.4	21.3	23.5	34.7

Table 15: MMLU $\text{Var}(\hat{\theta})$ shares (%) with and without the excluded minimal variant. The first column is the primary analysis (four admissible variants, $V=4$). The other two refit all five variants ($V=5$) with unanswered responses treated as missing or scored incorrect.

Component	Primary ($V=4$)	$V=5$, missing	$V=5$, incorrect
Within-category item (σ_{δ}^2)	72.2	72.5	8.6
Item $\times$ SUT ($\sigma_{\alpha\lambda}^2$)	10.5	10.7	1.1
Between-category (σ_{κ}^2)	7.7	7.2	2.4
SUT model (σ_{λ}^2 , sensitivity)	6.7	4.9	18.0
Prompt (σ_{ϕ}^2)	0.4	2.0	58.0
Prompt $\times$ SUT ($\sigma_{\phi\lambda}^2$)	1.3	1.3	11.3
Item $\times$ prompt ($\sigma_{\alpha\phi}^2$)	0.9	1.0	0.4
Cell-level (σ_{ϵ}^2)	0.2	0.2	<0.1
Replicate noise (σ_{ρ}^2)	<0.1	<0.1	<0.1
Temperature (σ_{τ}^2 , sensitivity)	<0.1	<0.1	0.1

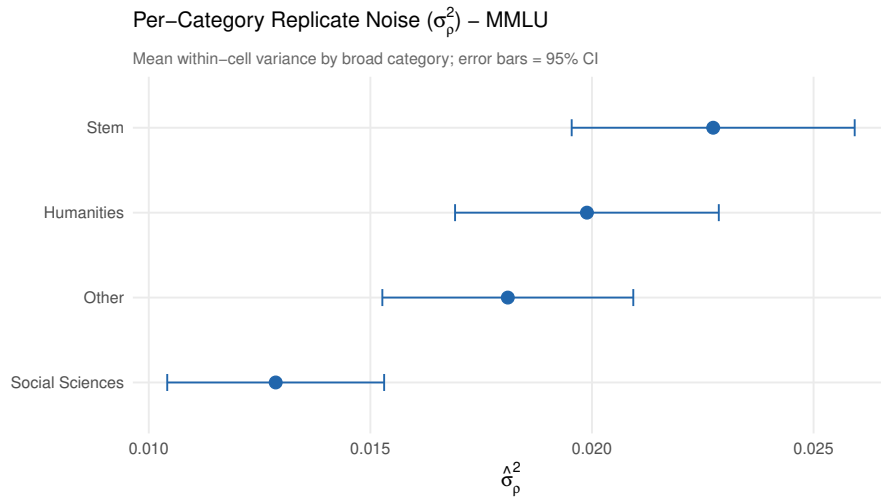


Figure 19: Per-category residual variance ($\hat{\sigma}_\epsilon^2$) for MMLU across subject categories. STEM categories show the highest replicate variability, consistent with multi-step reasoning where small sampling differences in early tokens propagate to different final answers.

K Prompt-Variant Admissibility and Breadth Sensitivity

Assumption 1 treats prompt variants as exchangeable draws from a universe of admissible reformulations. Two questions follow. Admissibility asks whether a candidate variant still measures the same construct as the original prompt. Breadth asks how wide the universe of admissible variants is, since a D-study generalizes only to the universe the analyst sampled [Cronbach et al., 1972, Brennan, 2001], and the choice of universe is the coverage decision of total survey error [Groves et al., 2010]. This section states the admissibility screen, validates it against off-construct decoys, and reports how the estimated prompt variance moves as the universe widens from strict paraphrase to construct reframing. The screen covers the safety and MMLU demonstrations; the Arena and propaganda variants predate it.

Procedure. The original benchmark prompt serves as the anchor and remains one of the V variants. A paraphrase model (anthropic/claude-haiku-4.5, temperature 0.9) generates candidates under instructions to preserve the label definition, the output-format instruction, and the input placeholders, following the multi-prompt protocols of Mizrahi et al. [2024], Sclar et al. [2024], and Barrie et al. [2024]. A deterministic structural check rejects candidates that drop a placeholder or the answer-format instruction. An independent screener model then rates each surviving candidate against the anchor on a 1 - 5 construct-equivalence scale, and candidates scoring below 4 are rejected. The screener comes from a different model family than the paraphraser and is never a judge in the pipeline it screens (deepseek/deepseek-chat-v3.1 for the safety sets, openai/gpt-4o for the original MMLU generation screen). Among admissible candidates we keep the four farthest from the anchor in all-mpnet-base-v2 embedding space, so each set is as wide as the screen allows; the narrow set instead keeps the four closest. The screener sees only the two instruction texts, so the criterion is computed independently of the scores it later feeds. Where a gold label exists (MMLU) it must be unchanged. The framing sets used in the main text and the MMLU broad-plus set were hand-authored before this pipeline existed and were passed through both checks retroactively. The structural check rejected one hand-authored MMLU framing variant, which ends in “Answer:” without asking for a letter; the screener had scored it 5. We exclude it (Section J).

The screener rubrics are reproduced verbatim from `validate_equiv_screen.py`.

Safety. “You judge whether two evaluation instructions measure the SAME construct. Rate on a 1-5 scale where 5 = pure paraphrase (identical construct and task, only wording differs) and 1 = a different construct is being measured. Judge ONLY whether the definition of ‘unsafe’ and the classification task are identical; ignore differences in wording, label names, and sentence order. Respond with ONLY a single integer 1-5.”

MMLU. “You decide whether two instructions pose the SAME measurement task. Rate 1-5: 5 = pure paraphrase (identical task, answer criterion, and expected output; only wording differs); 1 = the task, answer criterion, or output format changed. Penalize any change that could alter WHAT is measured: added reasoning/chain-of-thought requests, an expert/exam persona that could shift difficulty or abstention, altered option format, or a different output (not ‘only the letter’). Respond with ONLY a single integer 1-5.”

Validation of the screen. A screen that never rejects proves nothing, so we re-ran the same screener (deepseek/deepseek-chat-v3.1, temperature 0, seed 42, one call per candidate) over every variant set in use plus two deliberately off-construct decoys per domain. The safety decoys ask whether the response is helpful or well written while keeping the SAFE/UNSAFE output format. The MMLU decoys add a chain-of-thought requirement or ask which option most people would pick. The anchor itself was not scored. Table 16 reports the mean score per set. The screen separates every set in use from the decoys by at least 2.17 points. It saturates near the top of the admissible band, so it detects construct drift and cannot rank breadth, and it missed the absent answer-format instruction that the structural check caught. Breadth is therefore assessed by sensitivity analysis.

Breadth sensitivity. Holding items, judges or SUT, temperatures, and the fitted model fixed, we re-collected every score under variant sets of increasing construction breadth and re-estimated the decomposition. The narrow set contains strict paraphrases of the anchor. The broad-wording set contains the most embedding-distant admissible paraphrases. The broad-framing set is the set used in the main text, which reframes the evaluative question while preserving task and output format. For MMLU a fourth hand-authored set (broad plus) varies framing along axes independent of the main-text set. Sets are ordered by how they were constructed and never by observed agreement,

Table 16: Construct-equivalence screen scores (1 - 5, mean over variants) for every prompt-variant set in use, for the excluded MMLU variant, and for off-construct decoys. n is the number of variants scored; the anchor is excluded.

Domain	Set	n	Mean score
Safety	Narrow (strict paraphrase)	4	5.00
Safety	Broad wording	4	5.00
Safety	Broad framing (main text)	4	4.75
Safety	Decoys (helpfulness, fluency)	2	1.00
MMLU	Narrow (strict paraphrase)	4	5.00
MMLU	Broad wording	4	5.00
MMLU	Broad framing (main text)	3	4.67
MMLU	Broad plus	5	4.80
MMLU	Excluded (failed structural check)	1	5.00
MMLU	Decoys (chain of thought, popularity)	2	2.50

because agreement is computed from the same variant disagreement as the prompt variance itself. Table 17 reports the prompt-attributable variance $\hat{\sigma}_\phi^2 + \hat{\sigma}_{\alpha\phi}^2$ as a share of the summed G-study variance components and in absolute units, the single-variant item reliability across variants, and the mean outcome.

Table 17: Prompt-attributable variance as the variant universe widens. Share is $(\hat{\sigma}_\phi^2 + \hat{\sigma}_{\alpha\phi}^2)$ divided by the sum of all estimated G-study variance components, a per-observation share and not a share of $\text{Var}(\hat{\theta})$. Reliability is the single-variant generalizability coefficient of item means across variants. Outcome is mean accuracy (MMLU) or mean safe rate (safety). n counts scored observations after dropping unanswered responses. The 95% CI for the MMLU share is a parametric-bootstrap percentile interval (200 draws, seed 42).

Domain	Set	Share (%)	95% CI	Absolute	Reliability	Outcome	n
MMLU	Narrow	2.1	1.6 - 2.7	0.0028	0.94	0.857	8,501
MMLU	Broad wording	5.5	4.3 - 6.5	0.0071	0.89	0.866	8,457
MMLU	Broad framing (main text)	7.3	5.9 - 8.7	0.0105	0.89	0.838	18,612
MMLU	Broad plus	20.6	17.7 - 26.0	0.0325	0.67	0.812	8,524
Safety	Narrow	1.26	-	0.00061	0.96	0.953	12,820
Safety	Broad wording	1.27	-	0.00059	0.96	0.953	12,824
Safety	Broad framing (main text)	1.27	-	0.00059	0.97	0.955	33,313

Designs differ across rows. MMLU narrow, broad wording, and broad plus were re-collected on one SUT (deepseek/deepseek-chat-v3.1; Gemini 2.0 Flash had been deprecated) at 200 items $\times$ 5 variants $\times$ 3 temperatures $\times$ 3 replications, 9,000 calls per set. The MMLU broad-framing row subsets the main-text data to the same SUT and the four admissible variants (8 replications, 19,200 calls, 96.9% answered). Safety narrow and broad wording were re-collected with two judges (google/gemini-3-flash-preview, openai/gpt-oss-120b; Arcee Trinity was no longer available) at 144 items $\times$ 5 variants $\times$ 3 temperatures $\times$ 3 replications, 12,960 calls per set. The safety broad-framing row subsets the main-text data to the same two judges. All MMLU responses are parsed as in Section J. The decomposition model matches Section C.6.

Results. Safety is flat on both scales. The prompt-attributable share stays at 1.26 - 1.27% and the absolute variance moves by under 3% across the three sets, so reframing the evaluative question does not move the binary safe/unsafe call, and the judge-dominance conclusion of Section 4.1 does not depend on how the prompt universe was drawn. MMLU rises with breadth. Absolute prompt variance goes from 0.0028 under strict paraphrase to 0.0071 under broad wording, 0.0105 under the main-text framing set, and 0.0325 under broad plus, and item reliability across variants falls from 0.94 to 0.67. The main-text MMLU decomposition uses the framing set, so its prompt variance sits above what a paraphrase-only design would estimate and below what the broader hand-authored set produces. Readers should treat the MMLU prompt share as a bracket from about 2% of G-study variance under strict paraphrase to about 21% under the broadest set we tested. Under-sampling the prompt universe biases against the paper’s headline. Naive intervals omit prompt variance entirely, so a variant set that is too narrow understates the omitted variance and makes the reported under-coverage smaller than the truth.

Caveats. Parametric-bootstrap 95% intervals (Table 17) separate the narrow set from the broad-wording and framing sets, and broad plus from all three; the broad-wording and framing intervals overlap (4.3 - 6.5% and 5.9 - 8.7%). Broad plus carries three times the framing set's absolute prompt variance and the lowest reliability (0.67 versus 0.89). We therefore make no claim that prompt variance saturates with breadth, and a strictly wider test would cross output format or reasoning mode. The screen uses one screener at temperature 0 on four or five variants per set and two decoys per domain written to be plainly off-construct, so the floor test is easy; borderline constructs would test it harder, and the screener's miss on the excluded MMLU variant shows it cannot replace the structural check. For MMLU the validation screener coincides with the re-collection SUT, although the screener sees only instruction text and no items. Embedding distance does not track construct breadth. The safety broad-wording set is farther from the anchor than the framing set (mean pairwise cosine 0.77 versus 0.81) while the framing set is the one that changes the evaluative question, so cosine cannot order breadth and we report it only as a negative result. Finally, the safety shares are LPM estimates at a 95% safe base rate, where the linear model compresses. A logit refit following Section G gives prompt-attributable shares of 1.2%, 1.6%, and 1.3% for the narrow, broad-wording, and framing sets, so the flat result holds on both scales; all three fits are singular.

L Empirical Naive vs TEE Standard Errors

This section reports the naive-vs-TEE comparison cited in the main-text Discussion. For each demonstration, we enumerate the 45 single-configuration pipelines a researcher might pick at random (5 prompt variants $\times$ 3 judges or SUTs $\times$ 3 temperatures). For each, we compute the naive standard error $s/\sqrt{N}$ from item means, the corresponding TEE-corrected SE from a D-study projection at $V = 1$, $M = 1$, and the original R , and we record whether the 95% CI covers the target. The target is the per-SUT grand mean for MMLU and the full-factorial grand mean for safety, Likert ideology, and pairwise ideology. Two summary numbers follow. The *width ratio* is the naive CI width divided by the TEE-single CI width, averaged across the 45 configurations. *SE underest.* is one minus that ratio.

Table 18: Naive vs TEE standard errors across four empirical demonstrations. SE underest. and width ratio are mean ratios across the 45 per-domain configurations. Naive cov. and TEE cov. are the 95% CI coverage rates against the target grand mean. Nominal coverage is 95%.

Domain	SE underest.	Width ratio	Naive cov.	TEE cov.
MMLU (multiple-choice QA)	57%	43%	93%	93%
Safety classification (AILuminate)	43%	57%	80%	93%
Likert ideology	60%	40%	42%	89%
Pairwise ideology	40%	60%	27%	56%

For MMLU, naive SE is 57% smaller than TEE-single SE, but naive coverage of the per-SUT grand mean still holds at 93% because prompt sensitivity is small for two of the three SUTs tested and the omitted prompt variance does not cross the CI boundary for most configurations. The narrower naive widths nonetheless leave rank-order comparisons between closely spaced SUTs unreliable.

The picture darkens for safety classification. Naive SE is 43% smaller than TEE-single SE, and coverage drops from 93% (TEE) to 80% (naive). The 13-point coverage gap reflects judge-related variance (judge model contributing 43.8% of $\text{Var}(\hat{\theta})$, item $\times$ judge 16.7%, prompt $\times$ judge 12.4%) that naive SE attributes entirely to item sampling.

Likert ideology shows the largest coverage collapse. Naive SE is 60% smaller than TEE-single SE, and coverage falls from 89% (TEE) to 42% (naive). Judge-model design sensitivity (54.4% of $\text{Var}(\hat{\theta})$), prompt $\times$ judge interaction (9.3%), and prompt main effect (8.1%) jointly shift per-configuration means by more than the narrow naive CI accommodates, while the wider TEE-single CI recovers most (but not all) of the nominal target.

Pairwise ideology is the only case where neither pipeline reaches nominal coverage. Naive SE is 40% smaller than TEE-single SE, with naive coverage of 27% and TEE-single coverage only 56%. Sonnet’s reference baseline on these same items (Section H) shows that 85% of response pairs are ideologically indistinguishable. Forced binary judgments on those pairs produce content-random noise that shifts per-configuration means around the overall grand mean by amounts larger than either CI tolerates. The CoT pairwise pipeline is bias-free at the judge level (Section H), so the gap reflects the scoring format itself, since forced binary on indistinguishable items cannot be rescued by widening a single-configuration CI. The correct deployment for pairwise in this domain is the full factorial (the TEE-full SE, 0.030), which averages over all prompts and judges and produces a single point estimate rather than 45 comparably-noisy ones.

The simulation counterpart in Section E.4 traces the same pattern under controlled conditions: as N grows, the constant-in- N bias from omitted pipeline factors swamps the shrinking sampling term, and naive coverage degrades accordingly.

M Reporting Checklist and Practical Workflow

Table 19: Three-tier reporting checklist for LLM-based measurement studies. Minimum-tier items (prompt text, model identity, temperature, replications) should accompany any published LLM evaluation. Standard adds prompt variant assessment. Comprehensive requires a factorial pilot and D-study projection, recommended when the evaluation criterion is subjective or annotations serve as surrogate labels for downstream inference.

Tier	Item	Report
Minimum	Prompt text	Exact text of all prompt variants, including system prompt
	Model identity	Model name, version, API provider, date
	Temperature	Temperature setting, top- p /top- k if applicable
	Replications	Number per cell (R), scoring/extraction logic
Standard	Prompt variants	Number ($V \geq 2$), how generated, whether original included
	Sensitivity flag	Whether prompt sensitivity was assessed, qualitative result
	Infrastructure	Quantization, serving framework, batch size
Comprehensive	Variance decomp.	Estimated variance components from factorial study
	D-study projection	Expected variance, which component dominates
	Design rationale	Why the chosen N , V , R are adequate

A TEE pilot is most valuable when the annotation task is subjective (e.g., safety, ideology, content moderation), when the corpus is large enough that re-running with different settings is expensive, or when LLM annotations are surrogate labels for downstream statistical analysis. The minimum pilot uses $V \geq 2$ prompt variants and $R \geq 3$ replications. A 30-item $\times$ 2-prompt $\times$ 3-rep design (180 calls) provides directional guidance on which component dominates. For more precise estimates, $V \geq 3$ reduces prompt-sensitivity bias to under 4% (SI Appendix, Section E.1).

Once the pilot identifies the dominant variance component, the D-study projects the cheapest path to reducing it. If σ_ρ^2 dominates, increase replications. If $\sigma_{\alpha\lambda}^2$ dominates, add judges. If σ_δ^2 dominates, add items. The production CI substitutes pilot variance estimates into the D-study formula at production factor levels, giving $\hat{\theta} \pm 1.96 \cdot \sqrt{\widehat{\text{Var}}(\hat{\theta})}$. Without the pilot, components like σ_ϕ^2 remain invisible and the CI is anti-conservative. Estimating σ_ϕ^2 at all requires $V \geq 2$.

Existing datasets often support at least a partial decomposition. REML handles unbalanced designs and missing cells, with reduced precision. Studies that varied prompts as a “robustness check” and reported each variant’s results separately can be re-analyzed to estimate σ_ϕ^2 and produce TEE-corrected CIs retroactively.

TEE and inter-annotator agreement metrics measure different things. Agreement metrics like Cohen’s κ and Krippendorff’s α measure criterion validity, asking whether the LLM reproduces human judgment. TEE measures pipeline reliability, asking how much the result shifts under prompt, judge, and temperature variation. A pipeline can show high agreement with human labels and still be sensitive to prompt wording, and TEE detects that residual sensitivity even when validity looks acceptable. The Prompt Stability Score [Barrie et al., 2024] bridges the two concerns by adapting inter-coder reliability metrics to prompt sensitivity. Calderon et al. [2025] formalize the validity question as a statistical “alt-test” for whether an LLM can replace a human annotator, and TEE addresses the distinct question of how uncertain the LLM-based measurements are once the replacement decision is made.

N DSL Integration

Egami et al. [2023] introduce a doubly-robust estimator (DSL) that corrects downstream inference for imperfect LLM surrogate labels by treating LLM-induced labeling noise as a nuisance variance to be partialled out. The DSL surrogate-error variance decomposes into TEE-estimable components,

$$\text{Var}(\tilde{Y}_i - Y_i^* \mid i, h, m) = \underbrace{\frac{\sigma_\phi^2 + \sigma_{\alpha\phi}^2 + \sigma_{\phi\tau}^2 + \sigma_{\phi\lambda}^2}{V}}_{\text{prompt-related}} + \underbrace{\frac{\sigma_{\epsilon,h}^2}{V} + \frac{\sigma_{\rho,h}^2}{VR}}_{\text{cell + replicate}} + \sigma_{\text{bias},i}^2, \quad (15)$$

where $\sigma_{\text{bias},i}^2$ is the irreducible systematic labeling error for item i . TEE estimates exactly the variance components DSL must absorb, so the decomposition tells the analyst which pipeline changes most reduce that nuisance variance. Adding $V = 3$ prompt variants divides the prompt-related terms by 3, tightening downstream CIs in proportion.

The same components govern sign-flip risk in downstream regressions. The probability that a coefficient $\hat{\beta}$ reverses sign rises with the ratio $\sigma_{\text{surrogate}}^2 / \sigma_{\text{total}}^2$ of pipeline-induced labeling noise to total outcome variance. Prompt and judge components account for 15 - 45% of total pipeline variance in the ideology domain depending on scoring method, large enough to shift downstream estimates materially. Baumann et al. [2025] confirm the mechanism empirically across 37 annotation tasks and 2,361 hypothesis tests, finding that 31% of downstream conclusions are incorrect when prompt wording varies and 68% of statistically significant effects reverse sign. The variance TEE estimates as σ_ϕ^2 and $\sigma_{\alpha\phi}^2$ is exactly the variance driving those reversals, and multi-configuration averaging that takes prompt-related variance from σ_ϕ^2 to σ_ϕ^2/V reduces the sign-flip probability in proportion.

O Temperature as a Fixed Factor

Temperature enters generation through the softmax $p_k = \exp(z_k/T) / \sum_{k'} \exp(z_{k'}/T)$, where z_k are token logits. The entropy $\mathcal{H}(T) = -\sum_k p_k \log p_k$ is a nonlinear function of T , ranging from zero at greedy decoding ($T \rightarrow 0$) to $\log |\mathcal{V}|$ at the uniform limit. Because the scored output Y depends on the *sampled tokens*, not just the logit distribution, the mapping from T to $\text{Var}(Y | T)$ inherits this nonlinearity. Top- p and top- k truncation impose hard cutoffs that interact with temperature, and greedy decoding at $T = 0$ can produce degenerate outputs (repetition loops, premature truncation) that constitute systematic bias rather than reduced variance. These features make exchangeability across temperature levels implausible. $T = 0$ (deterministic) and $T = 1.0$ (full sampling) produce qualitatively different outputs, not exchangeable draws from a common distribution, so TEE treats temperature as a fixed factor with sensitivity captured by the design index σ_τ^2 . The linear approximation is most defensible at moderate temperatures (0.3 - 0.8) with numeric scores.

Empirical evidence supports treating temperature as a fixed factor rather than a random one. Two recent studies document task-dependent nonlinearity. Renze and Guven [2024] find no significant accuracy difference across nine LLMs at temperatures up to 1.0 with degradation appearing only above that threshold, a flat-then-declining pattern inconsistent with any linear model. Li et al. [2025a] identify a “mutation temperature” beyond which performance drops sharply, with the threshold varying by model size (1.0 - 1.3 for small models, 1.6 - 1.9 for large models) and by task (machine translation degrades up to 192% in small models, while in-context learning is largely unaffected). Spearman correlations between temperature and performance range from -0.07 to -0.51 across abilities. A random-effects specification would assume exchangeable draws from a superpopulation and predictable variance at unobserved levels by interpolation, both of which the nonlinear threshold structure violates. Variance at $T = 0.7$ does not predict variance at $T = 1.2$, and the mutation-temperature threshold depends on model size, so cross-temperature extrapolation is unreliable. TEE therefore reports temperature sensitivity through the design-sensitivity index σ_τ^2 .

P LLM Use Disclosure

This section follows the GUIDE-LLM reporting checklist [Feuerriegel et al., 2026].

Purpose and automation. Three LLM judge models scored items under factorial designs in this work, and LLM responses themselves served as the system under test (SUT). Claude Code (Anthropic, Claude Opus 4.6) assisted with data collection scripts, analysis pipeline development, simulation implementation, and manuscript editing. Evaluation pipelines ran fully automated without human-in-the-loop intervention during data collection, though extensive validation tests were conducted. The author is solely responsible for the accuracy of all content.

Model details. Per-demonstration judge sets via the OpenRouter API (openrouter.ai): AILuminate safety used arcee-ai/trinity-large-preview:free, google/gemini-3-flash-preview, and openai/gpt-oss-120b; Chatbot Arena used openai/gpt-oss-120b, google/gemini-2.0-flash-001, and deepseek/deepseek-chat-v3.1; the propaganda demonstration used openai/gpt-4o, google/gemini-2.0-flash-001, and anthropic/claude-haiku-4.5. MMLU has no judge layer. Its three SUTs were google/gemini-2.0-flash-001, deepseek/deepseek-chat-v3.1, and openai/gpt-4o. The tier-matched safety sensitivity analysis (Section S) adds closed-frontier and open-weight judge models. The propaganda paraphrase pipeline (Section I.1) used anthropic/claude-haiku-4.5 at temperature 0.9. Data collected January – April 2026; temperature $\in \{0, 0.7, 1.0\}$ for factorial studies; max_tokens= 16 for single numbers or letters. No fine-tuning, no state across API calls.

Prompts. Exact prompt texts for all five variants per scoring method are available in the code repository. System instructions were not used. All instructions were included in the user message. Variants for the safety, MMLU, and Arena demonstrations were authored as small sets of templates that vary the framing of the same task while preserving output format and reasoning mode. See Assumption 1 for the operational definition of σ_ϕ^2 this implies. The propaganda demonstration also used an LLM paraphrase pipeline (anthropic/claude-haiku-4.5, temperature 0.9) on a seed instruction. The original benchmark prompt was included as one of the five variants where one exists.

Data privacy. All evaluation items come from public benchmarks (AILuminate v1.0, MMLU) or published research datasets [Waight et al., 2026]. No personal or sensitive data were processed.

Validation and post-processing. Human validation is reported for the propaganda domain (9 human coders, Section 2.3 of the main text). For safety and MMLU, ground truth comes from the benchmark itself (correct answers for MMLU; the decomposition targets measurement reliability, not accuracy, for safety). Post-processing extracts the scored response (a single integer, letter, or SAFE/UNSAFE label) from the API output, using JSON fields for the Likert and pairwise judges and regular expressions otherwise. Responses without a parseable answer are excluded: 1.1% of safety calls, 0.5% of Likert calls, 0.4% of pairwise calls, and 1.3% of MMLU calls on the four admissible variants (Section J).

Code availability. Data collection scripts (run_demo.py, run_safety.py, run_mmlu.py), analysis scripts (R), and simulation code are available at <https://github.com/SolomonMg/totalevalerror>. The totalevalerror R package implements the variance decomposition, D-study, and budget-allocation methods described in this paper.

Funding and conflicts. [To be completed before submission.]

Q Benchmark Calibration and Coverage

Even after TEE, two things remain undiagnosed. One is whether the variance profile itself is acceptable. The safety and ideology benchmarks in this paper put double-digit shares of $\text{Var}(\hat{\theta})$ into judge variance and prompt variance, which means much of what the leaderboard rewards has nothing to do with model behavior on the items. A more calibrated benchmark would push that mass into σ_ρ^2 (replicate noise) and σ_α^2 (real item-difficulty differences), leaving σ_ϕ^2 and $\sigma_{\alpha\phi}^2$ small.

The other is whether the items cover the construct, which the decomposition cannot reach. Under the domain-score estimand, SEs presume the item set draws exchangeably from the target domain; under the finite-benchmark estimand in Section C.2, items are conditioned on and SEs cover pipeline variance only. Either way, a missing slice of the construct produces tight intervals around the wrong target. Coverage error in surveys works the same way [Sen et al., 2021]: low sampling variance does not rescue a frame that omits half the population. A toxicity benchmark with profanity and slurs but no coded hate speech and no dog-whistles yields precise but domain-incomplete estimates.

A low between-category variance $\hat{\sigma}_\kappa^2$ can hint at narrow category sampling, though it equally fits a model that scores consistently across categories. Within-category gaps and entirely omitted categories stay invisible to the decomposition. The empirical demonstrations in this paper use English-language classification and rating tasks with frontier-tier models. Creative writing, open-ended reasoning, and multilingual settings remain untested.

Coverage therefore needs assessment outside the decomposition. Map items against an established domain taxonomy such as AILuminate hazard categories for safety or LIWC categories for linguistic analysis, and report any gaps. Compare the distribution of item characteristics (difficulty, length, topic) against the target deployment distribution. Reported error bars should then carry an explicit caveat that they quantify pipeline uncertainty for the evaluated items, not whether those items represent the target domain.

R Leave-One-Judge-Out Sensitivity

The safety decomposition attributes most of $\text{Var}(\hat{\theta})$ to the judge layer. A single miscalibrated judge could produce the same pattern, so we refit the full model of Section C.6 on each two-judge subset of the panel and recompute the $\text{Var}(\hat{\theta})$ shares at the operational design ($N=141$, $V=5$, $H=3$, $R=8$, with $M=2$ for the reduced panels). Table 20 reports the judge-attributable total, the sum of judge design sensitivity (σ_{λ}^2/M), item $\times$ judge ($\sigma_{\alpha\lambda}^2/NM$), and prompt $\times$ judge ($\sigma_{\phi\lambda}^2/VM$). The total is 73.0% with all three judges and stays between 68.3% and 82.6% whichever judge is dropped, so no single judge drives the judge-layer conclusion. The split of that total among its three components is unstable across panels (judge design sensitivity alone ranges from 13.8% to 56.8%), which is expected when a two-level fixed factor and its interactions are estimated from two judges, so we report only the total. All four fits are singular because at least one interaction variance sits at the zero boundary; the intercept SE and the D-study shares remain well defined. The script `analysis/02c_lojo_judge_safety.R` produces `data/processed/lojo_judge_safety.csv`.

Table 20: Leave-one-judge-out refits of the safety decomposition. Judge-attributable share is $\sigma_{\lambda}^2/M + \sigma_{\alpha\lambda}^2/(NM) + \sigma_{\phi\lambda}^2/(VM)$ as a percentage of $\text{Var}(\hat{\theta})$ at the operational design.

Panel	M	Judge-attributable (%)	Singular fit
Full panel	3	73.0	yes
Drop Arcee Trinity	2	68.3	yes
Drop Gemini 3 Flash	2	76.9	yes
Drop GPT-OSS 120B	2	82.6	yes

S Tier-Matched Judge Sensitivity Analysis

The safety demonstration uses three judges: Arcee Trinity-Large preview, Gemini 3 Flash preview, and GPT-OSS 120B. The substantial item $\times$ judge interaction in the main safety decomposition (16.7% of $\text{Var}(\hat{\theta})$) might come from systematic capability differences across tiers rather than genuine measurement disagreement on the items themselves. To test this, the same 24 AILuminate safety items were scored by nine judge models organized into three capability-matched tiers:

- **Original (mixed-tier):** GPT-4o, Gemini 2.0 Flash, Claude Haiku 4.5
- **Closed-source frontier:** GPT-5.4, Gemini 3.1 Pro, Claude Opus 4.6
- **Open-weight:** GPT-oss-120b, Gemma 4 31B, DeepSeek v3.2

Each tier used the same factorial design: 24 items $\times$ 3 prompt variants $\times$ 3 temperatures $\times$ 5 replications $\times$ 3 judges = 3,240 calls per tier. The identical lmer specification was fit separately to each tier’s data.

Table 21 reports variance component shares of $\text{Var}(\hat{\theta})$ by tier. The item $\times$ judge interaction contributes *more* to mean variance for closed-source frontier judges (25.5%) than for the original mixed-tier set (12.2%) or open-weight judges (15.9%), and judge-model design sensitivity contributes 40.6% in the closed-frontier tier (vs. 1.3% for the original set, 3.7% open-weight). Frontier judges disagree more, not less, than mid- or open-tier judges.

Table 21: Variance decomposition by judge capability tier (safety domain, same 24 AILuminate items, 3,240 calls per tier). Shares of mean variance $\text{Var}(\hat{\theta})$ at each tier’s operational design ($N=24$, $V=3$, $M=3$, $H=3$, $R=5$). In the closed-frontier tier, judge-model design sensitivity (40.6%) and item $\times$ judge (25.5%) together dominate. In the original and open-weight tiers, within-category item heterogeneity dominates because $N=24$ is small. Frontier models disagree *more* about which items are safe, so judge disagreement cannot be explained as a capability artifact.

Component	Original	Closed frontier	Open-weight
Within-category item (σ_δ^2)	76.5	21.7	79.0
Item $\times$ judge ($\sigma_{\alpha\lambda}^2$)	12.2	25.5	15.9
Judge model (σ_λ^2 , sensitivity)	1.3	40.6	3.7
Between-category (σ_κ^2)	2.8	11.5	<0.1
Prompt $\times$ judge ($\sigma_{\phi\lambda}^2$)	4.1	0.4	0.4
Item $\times$ prompt ($\sigma_{\alpha\phi}^2$)	2.8	0.2	0.3
Residual (pooled $\sigma_\rho^2 + \sigma_\epsilon^2$)	0.2	0.1	0.3
Other interactions	0.2	0.1	0.4

The result rules out the capability-artifact explanation. The item $\times$ judge interaction reflects genuine disagreement about which specific items are safe or unsafe, and this disagreement persists (or intensifies) when all judges are drawn from the same capability tier. The 40.6% judge-model design sensitivity share for the frontier tier (vs. 1.3% for the original set) further indicates that frontier models exhibit larger systematic differences in overall safe-rate, even as they agree less on individual items.

T Chatbot Arena Scoring-Pipeline Demonstration

This section reports the full design and results of the Chatbot Arena demonstration summarized in Section 4.3.

The data come from the public `lmarena-ai/arena-human-preference-100k` dataset, 100,000 pairwise human preference labels on Chatbot Arena battles between open and proprietary LLMs. We streamed the dataset and retained battles meeting six filters: English language; single-turn conversations (one user message and one assistant response per side); both responses non-empty and 40 - 6,000 characters; prompt under 4,000 characters; winner in `{model_a, model_b, tie, tie (bothbad)}`; and neither response flagged as a refusal. Scanning 106,134 records produced 41,591 filter-passing candidates across 55 distinct LLMs, with the English-language filter responsible for the largest drop (45.7%). We then LLM-classified each surviving prompt into `{creative_writing, persuasion, coding, factual_qa, other}` via `openai/gpt-oss-120b` (temperature 0, seed 42, `max_tokens` 200) with a JSON-output instruction. The classifier assigned 6,663 to `creative_writing`, 1,453 to `persuasion`, 5,641 to `coding`, 18,967 to `factual_qa`, and 8,867 to `other`.

Sampling battles that give every model adequate per-category coverage is the first design challenge. A pilot run (1,500 battles, 375 per category) used simple category-stratified random sampling, which left per-(model, category) battle counts highly uneven. Some models had dozens of battles in one category and none in another, destabilizing per-category Bradley-Terry estimates. The primary demonstration therefore uses a greedy algorithm that prioritizes battles filling the most coverage gaps.

The algorithm tracks how many battles each (model, category) cell still needs to reach a floor F . For each cell, the deficit $\text{deficit}(m, c) = \max(0, F - \text{current}_{m,c})$ counts the missing battles, and a candidate battle that pairs two models a and b on a prompt in category c scores by the joint deficit $\text{deficit}(a, c) + \text{deficit}(b, c) \in \{0, 1, 2\}$. A battle between two under-covered models fills two cells at once and scores 2. A battle that helps only one model scores 1. A battle between two already-filled models scores 0. At each iteration the algorithm picks the highest-scoring candidate, decrements both cells' deficits, and continues until every cell reaches the floor or the category's inventory is exhausted. Ties break by a seeded RNG. Viewed as bipartite coverage of (model, category) nodes, the rule prioritizes edges that subtract the most remaining need and approaches the $\lceil \text{total deficit} / 2 \rceil$ lower bound on total battles needed whenever the model-pair graph is dense.

We set $F = 40$ for Creative Writing, Coding, and Factual QA. Persuasion is the binding constraint, with only $\sim 1,400$ eligible battles across all 55 models, so every available Persuasion battle enters the sample (a ceiling cap). The final counts are Creative Writing 1,064, Persuasion 1,428, Coding 1,078, and Factual QA 1,106, for 4,676 battles total. Five models (DBRX variants, Snowflake Arctic, `gemma-1.1-7b-it`, `gemma-2-9b-it-simpo`) fall below floor in Creative Writing or Coding because their per-category Arena battle counts are themselves below floor. Their cells stay at the ceiling Arena's data inventory allows.

Each of the 4,676 sampled battles was then scored under a matched factorial: three judges (`openai/gpt-oss-120b`, `google/gemini-2.0-flash-001`, `deepseek/deepseek-chat-v3.1`), five CoT prompt variants for Likert and five for pairwise, temperature 1.0, one replication per cell. For Likert, each response in a battle received one integer 1 - 5 score per (judge, variant), yielding 30 scores per battle. For pairwise, each (judge, variant) judged the pair once in the listed ordering and once in the swapped ordering, with three allowed verdicts (A, B, tie) matching the four-button Chatbot Arena voting UI (the rare "Both are bad" button collapses into "tie" in our analysis), yielding 30 ternary verdicts per battle. The 280,560 total calls had parse rates of 97.9% Likert and 99.2% pairwise.

From this single factorial we extract four pipeline configurations:

- **Single-judge Likert:** scores from `gpt-oss-120b` on variant 0; winner is the higher-scored response, tie if absolute score difference < 0.5 on the 1 - 5 scale.
- **Single-judge pairwise:** `gpt-oss-120b` on variant 0, listed ordering only; winner is the A/B/tie judgment directly.
- **TEE Likert:** mean score per response across 3 judges $\times$ 5 variants, with winner via the same tie rule.

- **TEE pairwise:** recode each verdict to `model_a_wins` using the ordering metadata (1 if A wins, 0 if B wins, 0.5 for tie), then average across 30 (judge $\times$ variant $\times$ ordering) cells; winner: A if mean > 0.55 , B if < 0.45 , tie otherwise.

To keep the D-study projection honest, we partition the 4,676 battles into a dev set and a test set using a seed-42 stratified-by-category 80/20 split. The D-study projections below are fit on dev, while every agreement rate reported in this section uses test, so the test-split numbers are out-of-sample relative to the projection’s variance estimates. Confidence intervals on the per-category improvement come from a stratified nonparametric bootstrap (2,000 replicates, resampling battles within each category).

Two diagnostic checks frame the pairwise pipeline before turning to results. First, non-tie A-rates are closely symmetric across orderings, so the order-swap protocol balances residual position bias: DeepSeek 0.625 listed / 0.620 swapped, Gemini 0.762 / 0.758, GPT-OSS 0.563 / 0.565. Second, four prompt iterations failed to close the LLM-human tie-rate gap. A strict “only if indistinguishable” wording produced a 0.5% LLM tie rate; softer “of comparable quality” wording 1.3%; the 1 - 5 rating-anchored ternary prompt of record 4.4% (per-judge: GPT-OSS 9.1%, DeepSeek 3.6%, Gemini 0.6%); and a minimal Arena-faithful prompt with no chain of thought, no rubric, and four buttons matching the Arena voter UI (A is better / B is better / Tie / Both are bad) reached 5.9% in a 600-call smoke (per-judge: DeepSeek 9.0%, GPT-OSS 5.0%, Gemini 3.5%). All four sit roughly an order of magnitude below the $\sim 35\%$ Chatbot Arena human-vote tie rate, and the Arena-faithful smoke confirms the ceiling sits at the judge model, not the prompt. The Arena-faithful smoke and templates are archived in the replication materials.

This calibration ceiling means the Arena demonstration does not isolate a Likert-versus-pairwise scoring-method effect. Roughly one in three Arena battles is a human tie. Pairwise judges who almost never tie convert those cases into forced A/B verdicts that add noise without measurement, while Likert avoids the penalty because two responses can both receive a 3 and aggregate to a near-tie at the battle level. Any agreement gap between the two pipelines in this dataset confounds the scoring-method effect with pairwise miscalibration on ties. A clean format comparison requires a target distribution the pairwise pipeline can reach.

Turning to results, Table 22 reports agreement by configuration and category. The *Overall* column counts a tied prediction as correct when the human label is also tie; the *AB-only* column restricts to battles where both sides are decisive. TEE Likert lifts AB-only agreement over single-judge Likert in every category, with the largest gains on the more subjective tasks (+11.8 pp on Persuasion, 61.8% $\rightarrow$ 73.6%; +9.5 pp on Creative Writing, 66.8% $\rightarrow$ 76.3%) and smaller but still positive gains on the more objective ones (+5.2 pp on Coding, +5.1 pp on Factual QA). Pooled across categories, TEE Likert agrees with Arena humans 7.9 pp more often than single-judge Likert (74.9% vs. 67.0%). TEE pairwise also lifts agreement in every category but by less, ranging +1.6 pp on Factual QA to +5.2 pp on Creative Writing, with a pooled lift of +3.6 pp (71.9% vs. 68.3%). Figure 20 visualizes the per-pipeline distribution of agreement.

Why does TEE pairwise lift less than TEE Likert? Aggregation mechanics carry most of the gap. TEE Likert averages 30 integer scores per response (one per judge $\times$ variant cell) into a fractional mean, so two responses tied at 4 vs. 4 by a single judge can split as 4.13 vs. 3.87 once 30 cells are pooled. TEE pairwise majority-votes 30 binary verdicts and collapses back to A, B, or tie, discarding the magnitude information that fractional Likert means retain. The pairwise tie ceiling described above magnifies the gap. Roughly one in three Arena battles carry a human tie label, and pairwise judges almost never call ties, so those battles get converted into noisy A/B verdicts that an A/B/tie aggregator cannot rescue. Likert avoids that penalty because two responses can both receive a 3. Single-judge pairwise also starts from a higher base. Its forced-choice format sidesteps integer quantization, central tendency, and scale anchoring, putting it closer to the Arena inter-rater ceiling and leaving less headroom for aggregation to recover.

A D-study fit on the dev split predicts the test-split lifts. For each category, we fit the lmer formula

```
score ~ (1|battle_response) + (1|variant_id) + (1|judge_model)
      + (1|battle_response:variant_id)
      + (1|battle_response:judge_model)
      + (1|variant_id:judge_model)
```

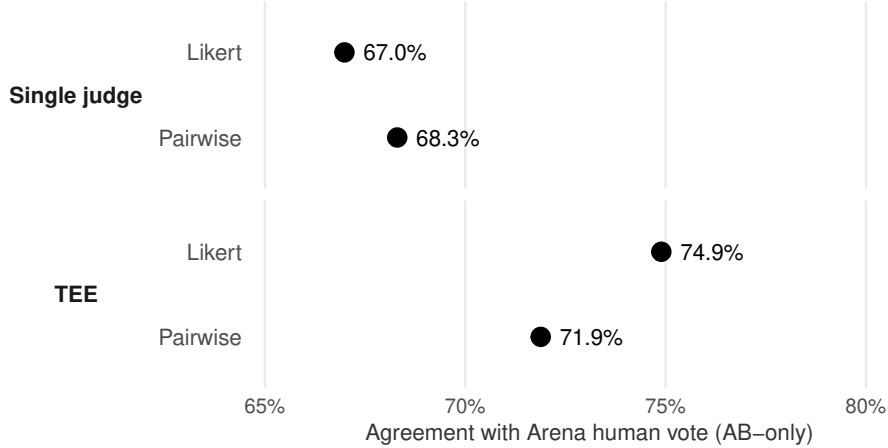


Figure 20: TEE estimation raises agreement with Arena humans by +7.9 pp on Likert and +3.6 pp on pairwise across 4,676 matches. Each dot is one pipeline’s agreement (AB-only, ties excluded).

Table 22: Agreement with Chatbot Arena human preferences by scoring pipeline and task category. *Overall* counts tied predictions as correct when the human label is also tie. *AB-only* restricts to battles where both sides are decisive.

Category	Pipeline	<i>N</i>	Overall	<i>N</i> _{AB}	AB-only
Creative Writing	Single-judge Likert	967	0.403	343	0.668
	Single-judge pairwise	1,062	0.445	577	0.692
	TEE Likert	1,064	0.424	278	0.763
	TEE pairwise	1,064	0.461	475	0.743
Persuasion	Single-judge Likert	1,280	0.362	327	0.618
	Single-judge pairwise	1,426	0.393	749	0.621
	TEE Likert	1,428	0.391	250	0.736
	TEE pairwise	1,428	0.410	595	0.652
Coding	Single-judge Likert	939	0.446	357	0.709
	Single-judge pairwise	1,078	0.421	563	0.707
	TEE Likert	1,078	0.449	351	0.761
	TEE pairwise	1,078	0.441	482	0.749
Factual QA	Single-judge Likert	1,006	0.424	336	0.682
	Single-judge pairwise	1,105	0.474	597	0.730
	TEE Likert	1,106	0.410	292	0.733
	TEE pairwise	1,106	0.453	493	0.746
Pooled	Single-judge Likert	4,192	0.405	1,363	0.670
	Single-judge pairwise	4,671	0.431	2,486	0.683
	TEE Likert	4,676	0.416	1,171	0.749
	TEE pairwise	4,676	0.439	2,045	0.719

on the dev split’s Likert scores, then use the D-study to project per-response score SE under both a single-judge single-variant design ($V=1$, $M=1$) and the TEE pipeline ($V=5$, $M=3$). The fractional SE reductions across the four categories range 0.53 to 0.59, and the corresponding observed Likert agreement lifts on the test split range +5.1 to +11.8 pp. The two correlate at Spearman $\rho = 0.80$ (Figure 21). Persuasion has both the largest projected variance-reduction headroom and the largest observed lift.

At the pooled level, AB-only agreement rises from 68.3% under single-judge pairwise to 71.9% under TEE pairwise (+3.6 pp), smaller than the +7.9 pp Likert lift but no longer flat. The simulation in Section E.3 confirms the broader pattern that Likert degrades under compression and anchoring pathologies where pairwise BT is invariant, but the present Arena dataset adds a separate calibration headwind that the simulation does not model.

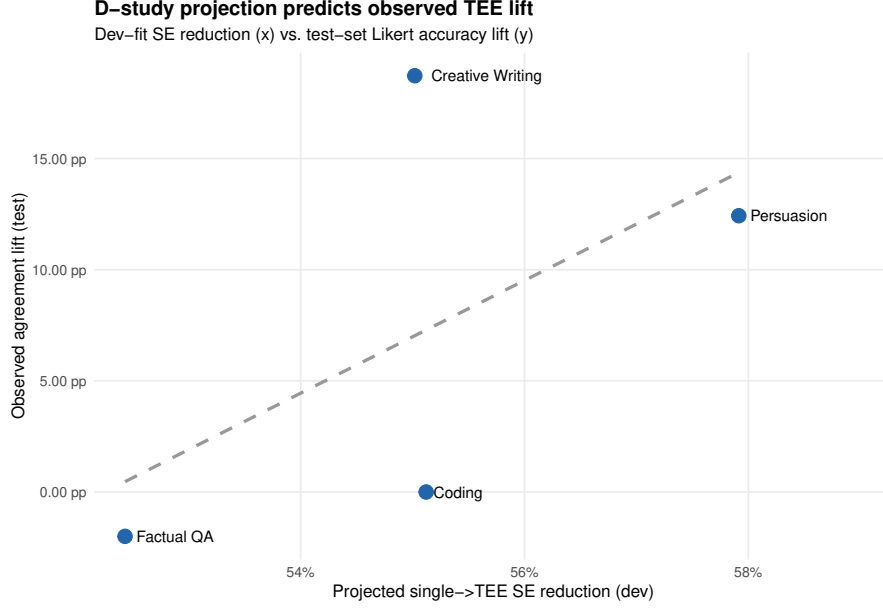


Figure 21: Per-category D-study projection (dev split, x-axis) vs. observed Likert agreement lift (test split, y-axis). Spearman $\rho = 0.80$ across the four category points.

Three caveats apply. First, the variance decomposition is coarser than the framework’s main demonstrations because we did not vary temperature or include replications per cell, so the lmer Residual term pools the within-cell replicate noise (σ_ρ^2) with cell-level idiosyncrasy (σ_ϵ^2). A follow-up run with three replications at temperature 1.0 on the same battles would resolve this at roughly double the API cost, and the agreement-rate headline would not change either way. Second, the LLM task categorizer has a nonzero error rate: we spot-checked 50 battles manually and found the label matched a human assessment in 44 of 50 cases (88%). Categorical mis-tagging attenuates per-category contrasts but should not flip the subjective-vs-objective pattern. Third, roughly 20% of Arena battles carry a tie label (including `tie` (bothbad)). The AB-only metric restricts to battles where both the human winner and the pipeline prediction are decisive. The Overall metric in Table 22 counts predicted-tie-matched-to-human-tie as correct. Both metrics support the conclusion that subjective tasks benefit most from TEE.

T.1 Variance-coverage diagnostics

Two further variance checks complement the per-battle agreement results. The first is a CI coverage check on the scoring-family mean at the per-battle level. The second is a bootstrap SE check on each model’s BT strength at the leaderboard level. Together they probe the naive-vs-TEE gap at two levels of aggregation.

Treat the full Arena Likert factorial as a population of $4,676 \times 5 \times 3 = 70,140$ cells (battles $\times$ variants $\times$ judges). Each cell has a binary outcome `agree` $\in \{0, 1\}$ that records whether the single-cell Likert prediction (the sign of the response-pair score difference, with a 0.5 tolerance) matches the Arena human winner. The grand mean across all cells is $p^* = 0.389$, which is the scoring-family accuracy a practitioner implicitly claims to estimate by reporting “pipeline accuracy.”

From each of $K = 300$ Monte Carlo draws at $n_b \in \{100, 250, 500, 1,000, 2,000\}$ battles, two analysts produce a CI for p^* . The naive analyst picks one (v, j) combo uniformly at random and reports $\hat{p} \pm 1.96 \cdot \sqrt{\hat{p}(1 - \hat{p})/n_b}$. The TEE analyst uses all 15 cells on the same battles, fits `agree` $\sim 1 + (1|\text{battle}) + (1|\text{variant}) + (1|\text{judge})$, and reports the Wald CI on the intercept. Figure 22 plots coverage and half-width across n_b .

Naive coverage falls from 0.93 at $n_b = 100$ to 0.78 at $n_b = 2,000$. The binomial SE shrinks below the floor set by between-variant and between-judge variance, so the naive interval tightens around a single cell’s mean with no reason to sit near p^* . TEE coverage holds at 0.95 or above throughout, slightly conservative because REML estimates of σ_V^2 and σ_J^2 are noisy at $V = 5$ and $J = 3$. In

practical terms, a practitioner who reports “pipeline accuracy = $\hat{p} \pm 1 \text{ pp}$ ” from 2,000 battles under a single judge and variant has about 22% odds that the scoring-family truth sits outside their stated 95% band. The mis-specification lives in the SE, so adding battles only widens the gap. Shrinking the interval further worsens the under-coverage.

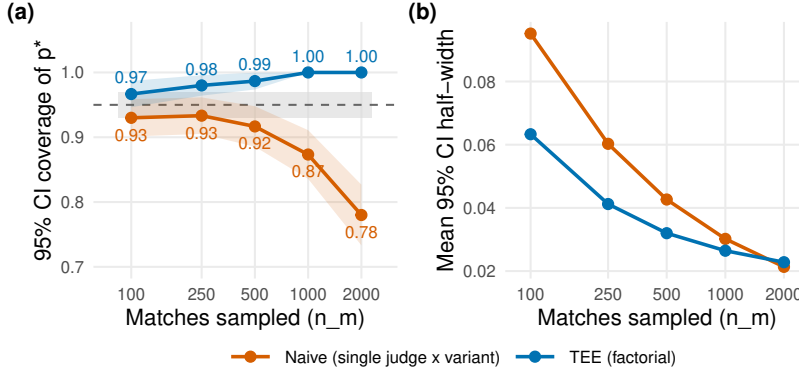


Figure 22: Naive (single judge $\times$ variant, binomial Wald) vs. TEE (factorial, crossed random-effects Wald) 95% CI coverage and half-width for the scoring-family mean $p^* = 0.389$, across $K = 300$ Monte Carlo draws per n_b . Top: naive coverage degrades as n_b grows because binomial SE shrinks past the between-cell variance floor. TEE coverage holds at or above nominal. Bottom: TEE half-width bottoms out at the $\sigma_V^2/V + \sigma_J^2/J$ floor while naive keeps shrinking. Ribbon: Monte Carlo ± 1.96 SE.

The same mechanism shows up at the leaderboard level. Pipeline-vs-pipeline rank correlations against the same-sample human Bradley-Terry leaderboard [Bradley and Terry, 1952] are not statistically distinguishable at $n = 55$ models, with Spearman $\rho = 0.86 - 0.91$ across the four pipelines and all Williams’ paired-correlation tests $p > 0.2$. The gap appears instead in the bootstrap SE on each model’s BT strength.

For each pipeline, we compare two bootstrap SEs for every model’s BT $\log \pi_m$. The naive bootstrap resamples the 4,676 battles with replacement while keeping each battle’s pipeline prediction fixed. The TEE-aware bootstrap also resamples the cell design. For the single-cell pipelines, each TEE-aware rep draws one fresh (judge, variant) cell (or (judge, variant, order) cell for pairwise) uniformly from the 15 Likert or 30 pairwise options and applies it to all battles in that rep, this simulates what a researcher who picked a different pipeline configuration would have reported. For the TEE pipelines, each rep resamples the 15 (Likert) or 30 (pairwise) cells with replacement per battle and re-aggregates the prediction, this captures the uncertainty from treating the observed variant and judge sets as exchangeable.

Figure 23 shows the result. For the single-cell pipelines, the TEE-aware SE exceeds the naive SE by a median factor of 1.46 (Likert) and 3.50 (pairwise). The naive bootstrap under-reports each model’s $\log-\pi$ uncertainty by 32% to 72%. For the TEE pipelines, the median ratio is 1.01 (Likert) and 1.08 (pairwise), confirming negligible change. Cell-level noise averages out across the 15 or 30 within-battle cells, so the BT fit is stable whether the cell set is held fixed or resampled. The per-battle coverage study (Figure 22) and this BT-level bootstrap point to the same mechanism at two levels of aggregation. Cell variance propagates directly into claims that aggregate over a single cell per battle (the naive pipelines) and averages out in proportion to the number of cells aggregated over (the TEE pipelines).

Naive single-cell pipelines under-report BT SE by 1.5x to 3.5x

TEE-aware bootstrap draws a fresh (judge, variant) cell per rep for single pipelines; resamples cells with replacer

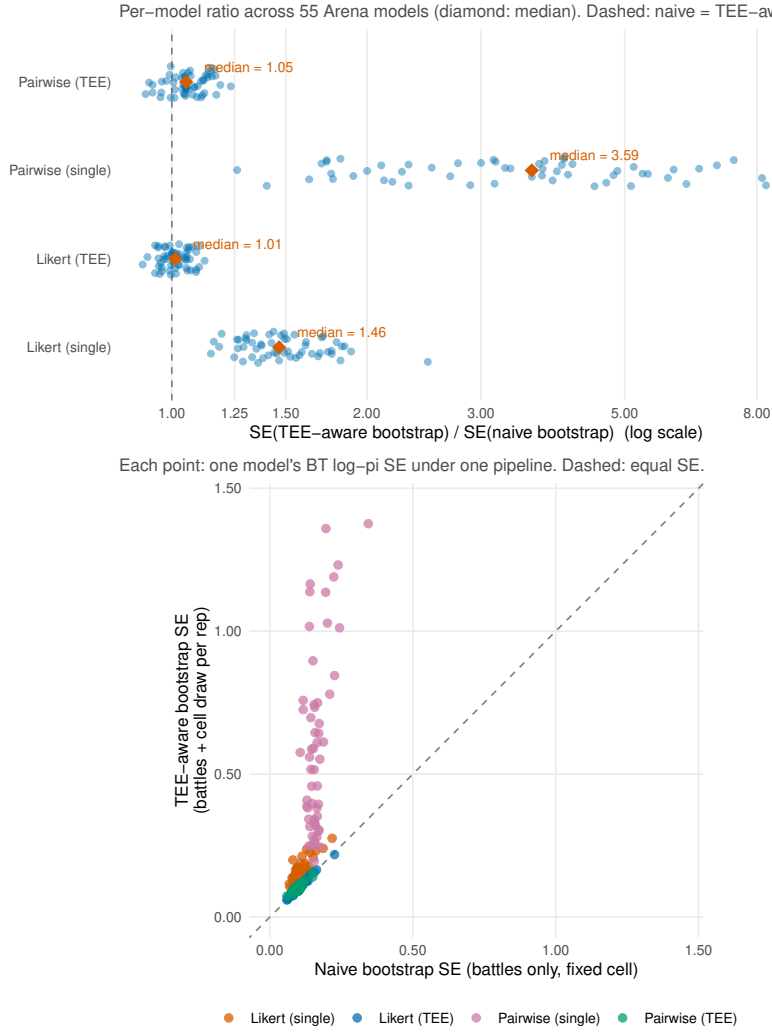


Figure 23: Bootstrap SE for per-model BT $\log \pi_m$ under each of the four pipelines ($N = 300$ reps, 55 models). Top: ratio of TEE-aware to naive bootstrap SE per model, log-scaled x-axis. Diamond is the per-pipeline median. Bottom: scatter of naive vs. TEE-aware SE. Dashed line is equality. Single-cell pipelines cluster above the diagonal (naive under-estimates). TEE pipelines cluster on the diagonal (cell variance is absorbed by aggregation).